1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

J95 MLB SKL EMERALD

LAST_MODIFICATION=Thu Feb 12 14:48:08 2015

REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
			2015-02-12

DRAWING
TITLE=J95_skl_am
ABBREV=DRAWING
LAST_MODIFIED=Thu Feb 12 14:48:08 2015

SCHEM,J95, mlb_skl_EM

DRAWING TITLE
SCHEM,MLB,SKYLAKE,EMERALD,J95

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DRAWING NUMBER 051-00673 SIZE D
REVISION 0.24.0
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DRAWING

8	7	6	5	4	3	2	1

BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
985-00098	PCBA,DEV,MLB_SKL,J95	DEVELOPMENT,J95_DEVEL
685-00062	PCBA,MLB_SKL,COMMON PARTS,J95	J95_COMMON
639-00957	PCBA,MLB SKL,EMERALD PROA,VRAM_HYNIX,2GB,J95	J95,MLB_CMNPTS,CPU:BEST,SSD:Y,GPU:EMERALD_PROA,FB:4G_HYNIX,EEEE:GHN4
639-00959	PCBA,MLB SKL,EMERALD PROA,VRAM_ELPIDA,2GB,J95	J95,MLB_CMNPTS,CPU:BEST,SSD:Y,GPU:EMERALD_PROA,FB:4G_ELPIDA,EEEE:GHN5
639-01039	PCBA,MLB SKL,95W,EMERALD PROA,VRAM_HYNIX,2GB,J95	J95,MLB_CMNPTS,CPU:CTO_2.6G,SSD:Y,GPU:EMERALD_PROA,FB:4G_HYNIX,EEEE:GKCT
639-01040	PCBA,MLB SKL,95W,EMERALD PROA,VRAM_ELPIDA,2GB,J95	J95,MLB_CMNPTS,CPU:CTO_2.6G,SSD:Y,GPU:EMERALD_PROA,FB:4G_ELPIDA,EEEE:GKCV

Bar Code Labels / EEEE #'s

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7896	1	LABEL,MLB,2D	EEEE_GHN4	CRITICAL	EEEE:GHN4
825-7896	1	LABEL,MLB,2D	EEEE_GHN5	CRITICAL	EEEE:GHN5
825-7896	1	LABEL,MLB,2D	EEEE_GHN4	CRITICAL	EEEE:GKCT
825-7896	1	LABEL,MLB,2D	EEEE_GHN5	CRITICAL	EEEE:GKCV

J95 SCHEMATIC / PCB #'S

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
051-00673	1	SCH,MLB SKL EM,J95	SCH1	CRITICAL	J95
820-00291	1	PCBF,MLB SKL EM,J95	PCB1	CRITICAL	J95
685-00062	1	PCBA,MLB SKL EM COMMON PARTS,J95	CMNPTS	CRITICAL	MLB_CMNPTS

BOM Groups

BOM GROUP	BOM OPTIONS
J95_COMMON	COMMON,VR_BULKCAP:CURRENT,ALTERNATE,J95_PROGPARTS,GPU_AM_EM,GPU_Emerald,XDP,SPEAKERID,FBVDDQ_DFLT:1V5,SMCREG:SUP
J95_PROGPARTS	SMC:PROG,BOOTROM:PROG,TBTROM:PROG,CIVROM:PROG,CAMROM:PROG
J95_DEVEL	XDP_CONN,T112,DEVEL_AUDIO

J95 ALTERNATES

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
377S0147	377S0126		ALL	USB Diode Array
197S0481	197S0480		ALL	25MHX PCH XTAL
197S0479	197S0478		ALL	12MHZ CAM/NXP XTAL
372S0186	372S0185		ALL	Alternate Temp Diode
107S00030	107S00029		ALL	Alt 5mOhm sense
107S0255	107S0240		ALL	Alt 1mOhm sense
341S00016	341S3912		U3990	IC,ENETROM,ADESTO,V1,15,J78
138S0681	138S0638		ALL	10uF Caps
138S0860	138S0775		ALL	Single-source 1uF 402
128S0398	128S0220		ALL	3.3V INPUT CAP
378S0391	378S0390		ALL	DEGUG LEDS
311S00008	311S0271		ALL	AND GATE
155S0830	155S0316		ALL	FERR BD,600 OHM,25,0.5A
376S00016	376S0644		ALL	MOSFET,P-CH,20V,1A
376S00001	376S0659		ALL	MOSFET,P-CH,20V
376S00015	376S0610		ALL	XSTR,FET,DL NCH,60V
138S1103	138S0719		ALL	4.7uF 0402
311S00004	311S0370		ALL	Config. Logic Gate
311S0250	311S0377		ALL	AND GATE
132S0447	132S0448		ALL	0.1uF 0201
138S0855	138S0750		ALL	4.7uF 0402
353S00095	353S3328		ALL	2:4 Diff MUX
311S00072	311S0657		ALL	Dual 2 INP AND GATE

CPUs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S00113	1	CPU,SKL,QH73,ES,P0,4/2,2.3,95W,LGA1151	CPU	CRITICAL	CPU:CTO
337S00122	1	CPU,SKL,QHQF,ES2,Q0,2.6G,95W,LGA1151	CPU	CRITICAL	CPU:CTO_2.6G
337S00114	1	CPU,SKL,QH8E,ES,P0,4/2,2.2,95W,LGA1151	CPU	CRITICAL	CPU:ULT
337S00121	1	CPU,SKL,QH8F,ES2,P0,4/2,2.2GHZ,65W,LGA1151	CPU	CRITICAL	CPU:BEST

CPU SOCKET

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
511S00002	1	SOCKET.MOLEX,LGA1151,H4	U0500	CRITICAL	

ASICs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S00124	1	IC,PCH-H,SKL,SUPRSKU,C0,QHPT,ES2,BG1151	U1100	CRITICAL	
338S1247	1	IC,TBT,FR-4C,A0,PRQ,CIO,SR1JC,FCBGA288	U2800	CRITICAL	
343S0616	1	IC,BCM57766A1,ENET&SD,8X8	U3900	CRITICAL	

GPU and VRAM

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S00077	1	IC,GPU,AMD,ES,EMERALD,PROA,128BIT,FCBGA1093	U8700	CRITICAL	GPU:EMERALD_PROA
333S00043	4	IC,HYN,GDDR5,4GBIT,64MX32,7GBPS,25NM	U9200,U9250,U9300,U9350	CRITICAL	FB:4G_HYNIX
333S00044	4	IC,ELP,GDDR5,4GBIT,64MX32,7GBPS,25NM	U9200,U9250,U9300,U9350	CRITICAL	FB:4G_ELPIDA

Programmable Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
341S00058	1	IC,EFI,V0174,J78	U5210	CRITICAL	BOOTROM:PROG
335S00006	1	IC,64 MBIT SPI SERIAL FLASH	U5210	CRITICAL	BOOTROM:BLANK
341S00262	1	IC,SMC-B1,EXTERNAL,VXXXX,POC-S,J78AM	U5000	CRITICAL	SMC:PROG
338S1214	1	IC,SMC12-B1,40MHZ/50DMIPS,MCU,157BGA	U5000	CRITICAL	SMC:BLANK
341S00152	1	IC,T29,EPROM,FR,VTBD,POC,J95	U2890	CRITICAL	TBTROM:PROG
335S0915	1	IC,SRL SPI FLASH ROM,4MBIT,50MHZ,USON8	U2890	CRITICAL	TBTROM:BLANK
341S3912	1	IC,ENET ROM,NUMONYX,V1.15,J16/J16G/J17	U3990	CRITICAL	CIVROM:PROG
335S0854	1	1-MBIT,SPI FLASH ROM SOIC, HF	U3990	CRITICAL	CIVROM:BLANK
341S3778	1	IC,CAMERA FLASH,V7230,J16/J17	U4202	CRITICAL	CAMROM:PROG
335S0852	1	IC,FLASH,SPI,1MBIT,3V3	U4202	CRITICAL	CAMROM:BLANK

SYNC_MASTER=J17_DINI		SYNC_DATE=03/15/2013
PAGE TITLE: BOM Configuration		
Apple Inc.	DRAWING NUMBER: 051-00673	SIZE: D
	REVISION: 0.24.0	
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	SHEET: 2 OF 93	

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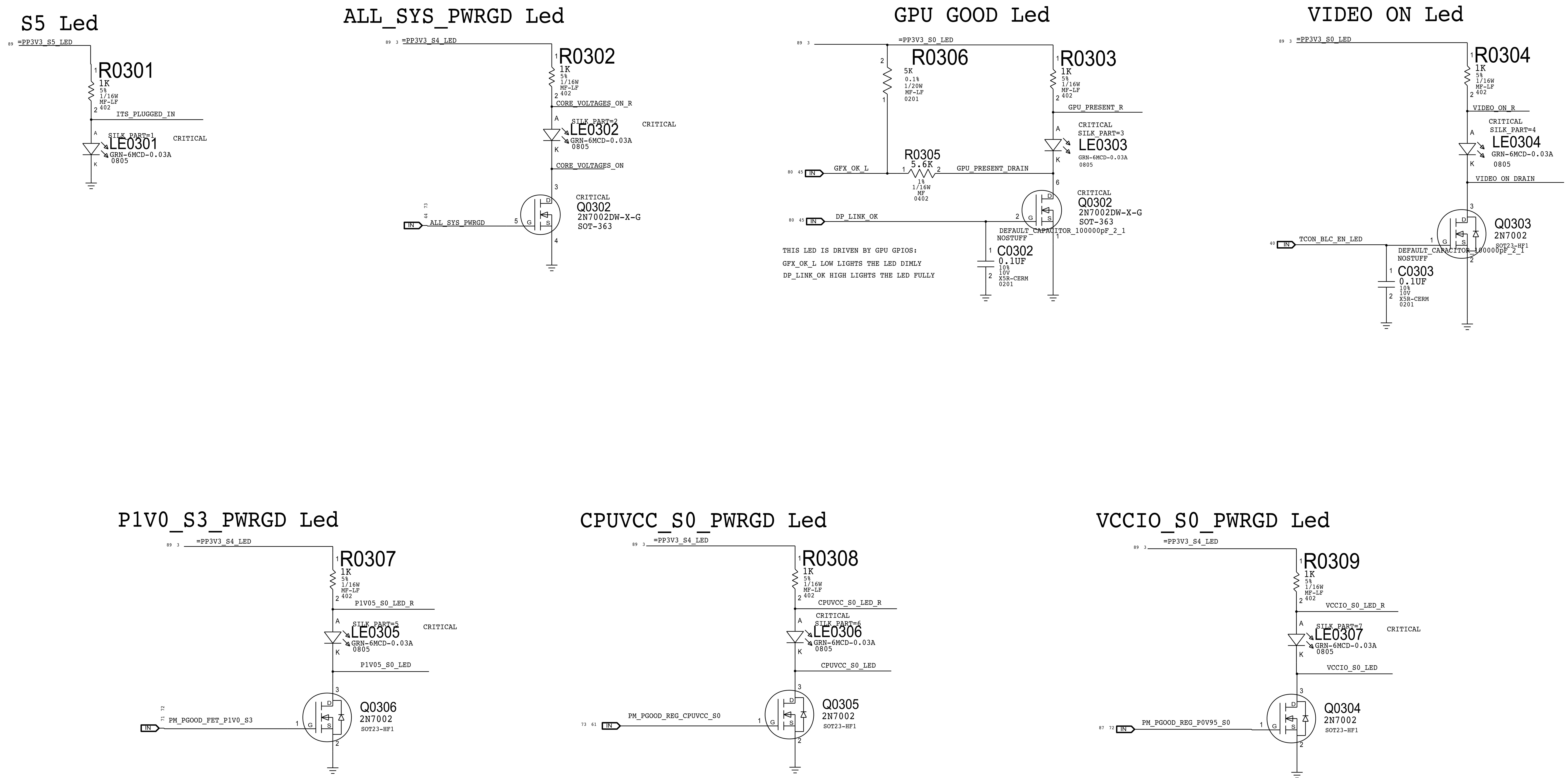

SYNC_MASTER=J17_MAX SYNC_DATE=02/11/2013

PAGE TITLE

DEBUG: LEDs

Apple Inc.

DRAWING NUMBER	051-00673
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CPU Heatsink
4MM PLATED HOLES (998-4158)

ZH0400 8P5R5-NSP
ZH0401 8P5R5-NSP
ZH0402 8P5R5-NSP
ZH0403 8P5R5-NSP

Display EMI Springs
APN: 870-00886

SF0401 SPRING-FINGER-6.26H SM
SF0402 SPRING-FINGER-6.26H SM
SF0403 SPRING-FINGER-6.26H SM

GPU HEATSINK MOUNTING FEATURES
(998-5013. PLATED HOLE, 3.2MM DIA, 6MM PAD TOP/BOT)

ZH0420 6P0R3P2-NSP
ZH0423 6P0R3P2-NSP
ZH0424 6P0R3P2-NSP
ZH0425 6P0R3P2-NSP

EMERALD EMI FENCE

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
860-02579	1	FENCE,GPU,EMERALD,90,J95	SF0404	CRITICAL	GPU_Emerald

Rear Cover
998-5014 (PLATED HOLES, 4MM DRILL, 8.5MM TOP, 8MM BOT)

ZH0413 8P5R4P0-8P0B-NSP
ZH0414 8P5R4P0-8P0B-NSP
ZH0416 8P5R4P0-8P0B-NSP
ZH0417 8P5R4P0-8P0B-NSP

Rear Cover
860-5674 (PCB STANDOFF)

ZH0415 STDOFF-7P14OD16P45H-TH-1.5-5.2A
ZH0418 STDOFF-7P14OD16P45H-TH-1.5-5.2A

T29 BUMPER
APN: 860-00201

NUT0401 2.80D1.2ID-2.6H-SM

HEATPIPE MTG HOLES
998-5527 (PLATED HOLES, 8MM DIA, 10MM PAD)

ZH0426 10.0R8.0

X238D WIRELESS CARD MTG HOLES
998-01489 (PLATED HOLES, 2.1MM INNER DIAMETER, 5.5MM PAD)

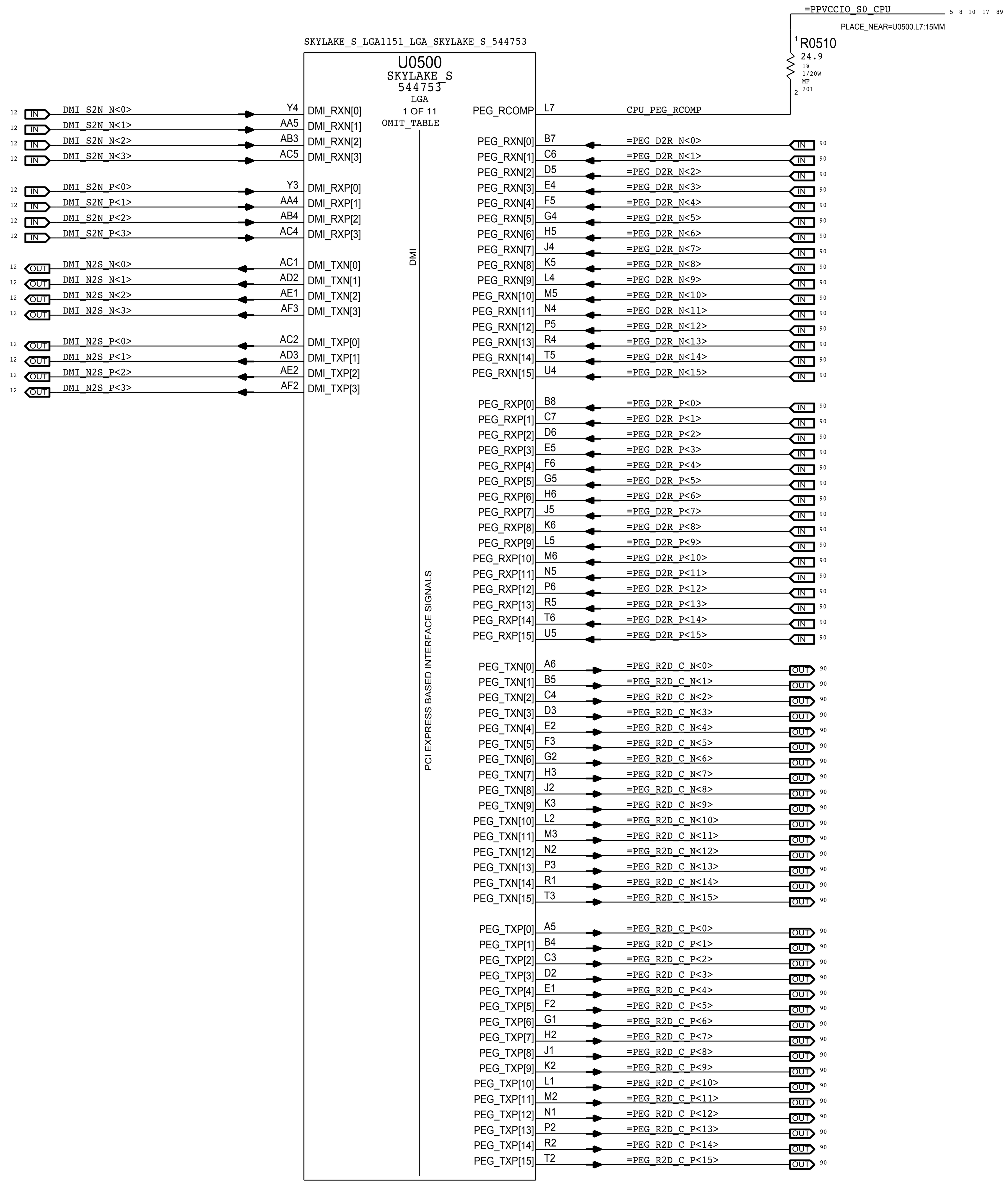
ZH0421 4P3R2P1-5P5B-NSP
ZH0422 4P3R2P1-5P5B-NSP

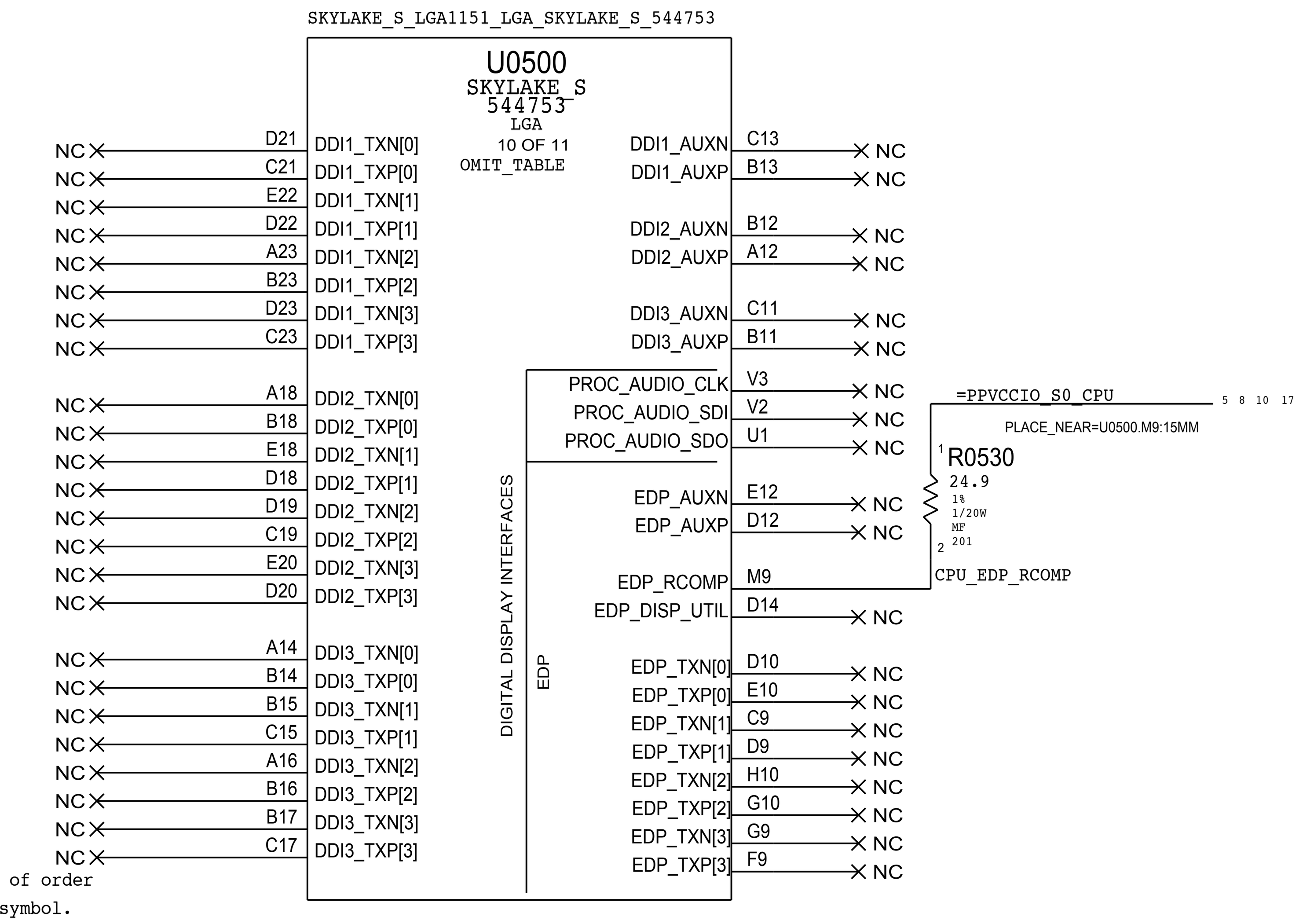
SSD STANDOFF
APN: 860-00198

CRITICAL
NUT0413 5.50D2.65ID-6.5H-SM

LOL Boss
860-5675 (PLATED HOLE, 8.41mm pad Top, 8mm pad Bot)

NUT0402 STDOFF-7.14OD10H-TH-1.5-5.2





Port D pins out of order
to match Intel symbol.

SYNC_MASTER=J78_MLB SYNC_DATE=06/30/2014

PAGE TITLE

CPU & CHIPSET: CPU DMI/PEG/FDI/RSVD

Apple Inc.

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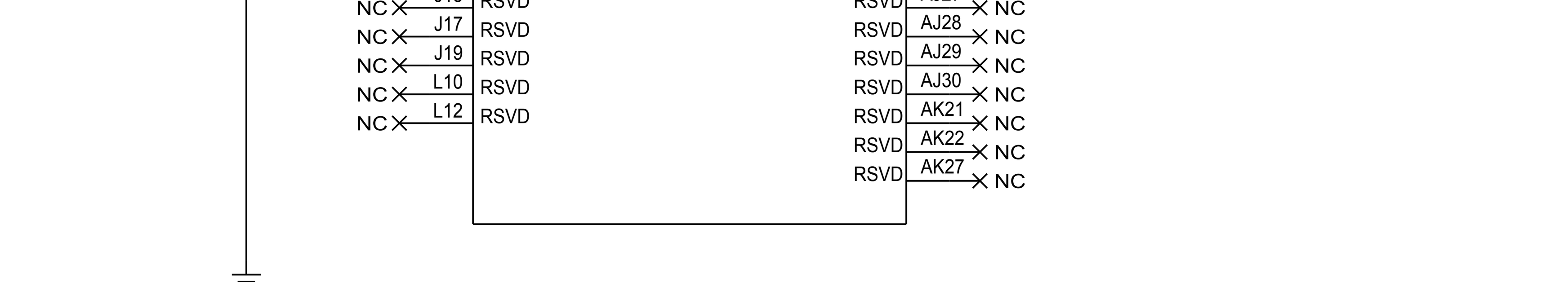

A

CFG [7] :PEG DEFER TRAINING	1 = (DEFAULT) IMMEDIATELY AFTER xxRESETB 0 = WAIT FOR BIOS
CFG [6:5] :PCIE BIFURCATION	11 = 1 X16 (default) 10 = 2 X8 01 = RSVD 00 = X8, X4, X4
CFG [4] :eDP ENABLE/DISABLE	1 = DISABLED(default) 0 = ENABLED
CFG [3] :PCIE x4 LANE REVERSAL	1 = NORMAL OPERATION(default) 0 = LANES REVERSED
CFG [2] :PCIE x16 LANE REVERSAL	1 = NORMAL OPERATION(default) 0 = LANES REVERSED

SYNC_MASTER=J78_MLB SYNC_DATE=06/30/2014

PAGE TITLE

CPU & CHIPSET: CPU Clock/Misc/JTAG/CFG

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SYNC_MASTER=J78_MLB SYNC_DATE=06/30/2014

PAGE TITLE

CPU & CHIPSET: DDR3L Interfaces

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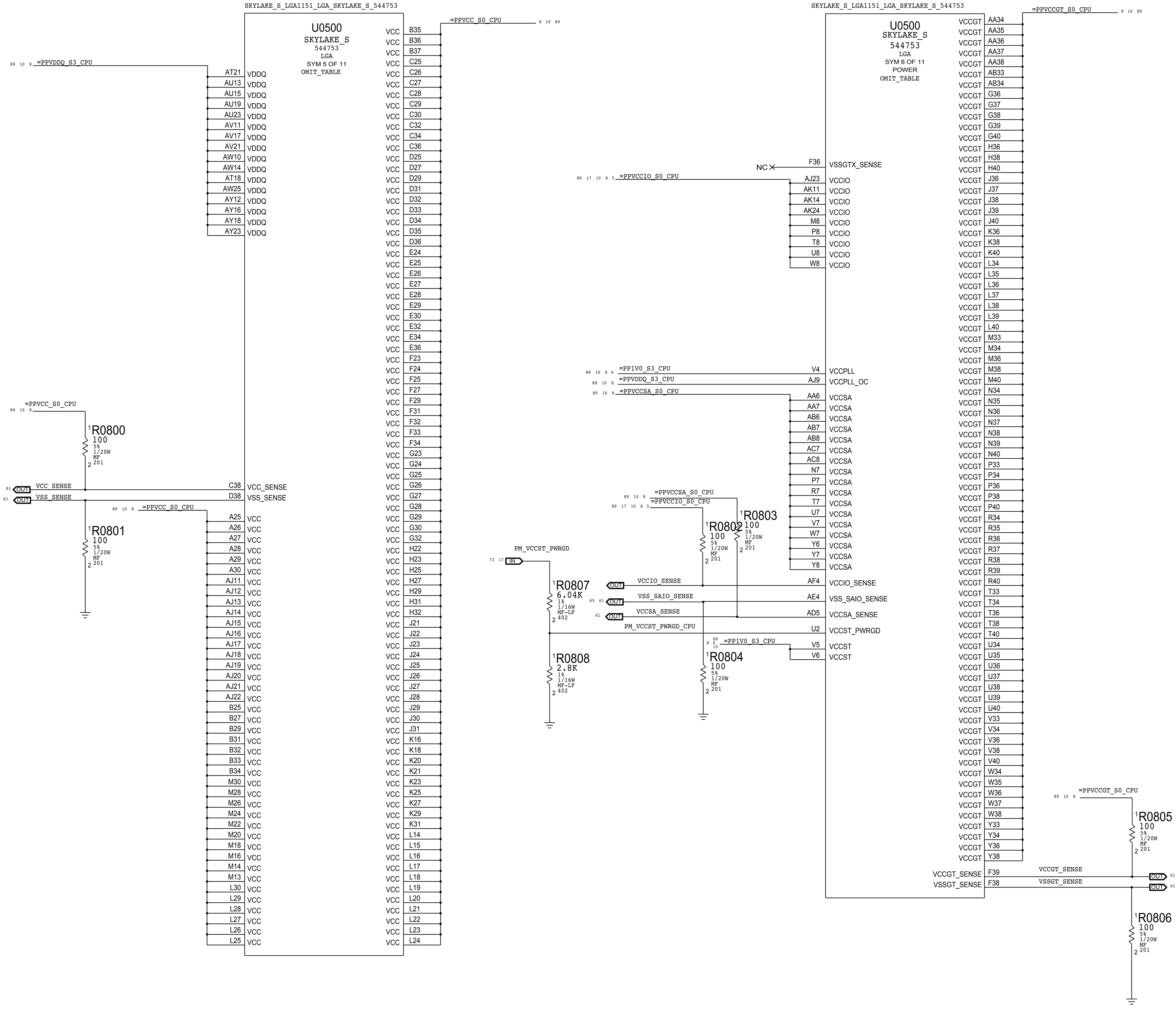

SYNC_MASTER=J78_MLB SYNC_DATE=06/30/2014

PAGE TITLE

CPU & CHIPSET: CPU Power

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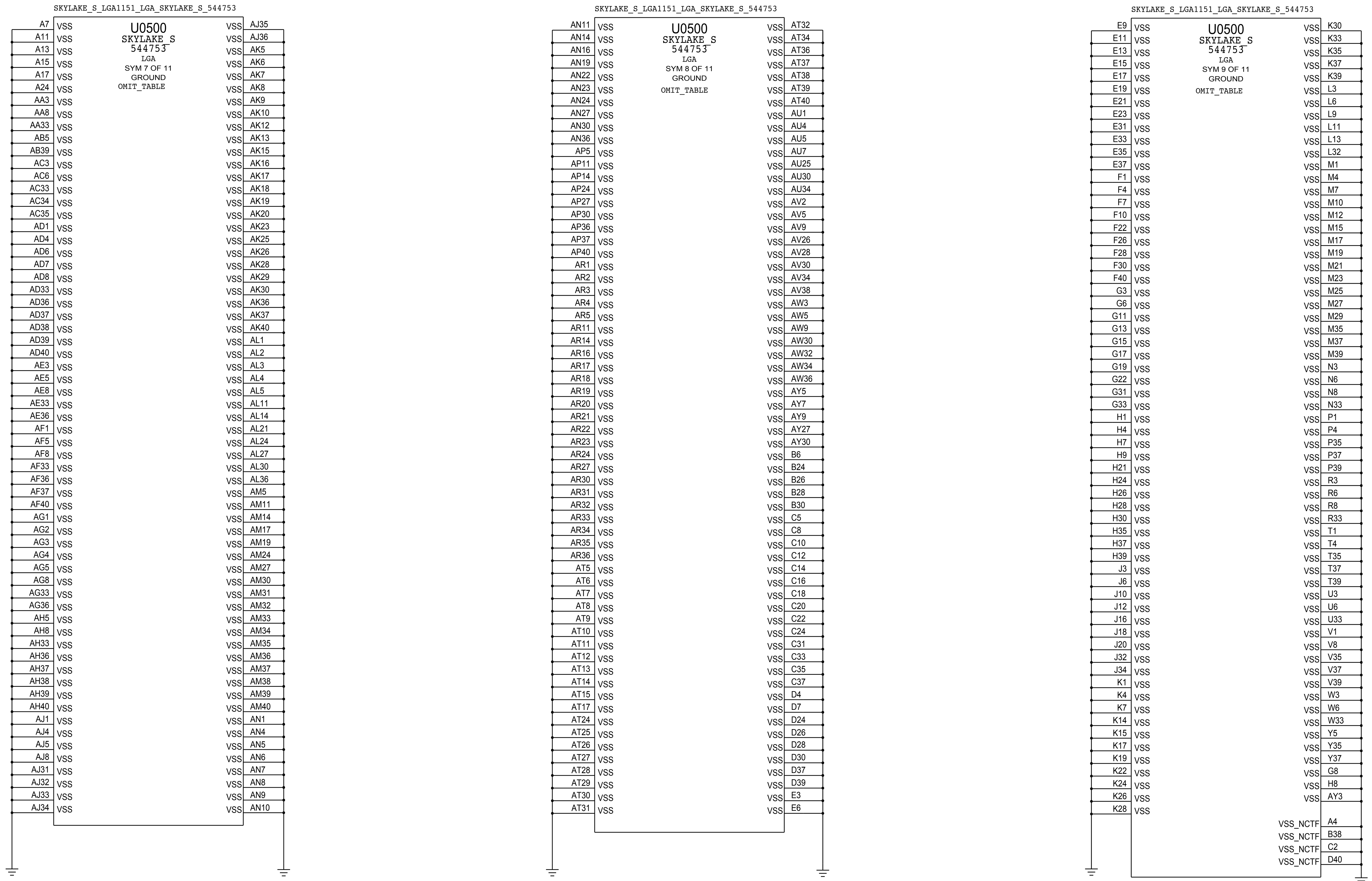

SYNC_MASTER=J78_MLB SYNC_DATE=06/30/2014

PAGE TITLE

CPU & CHIPSET: CPU Ground

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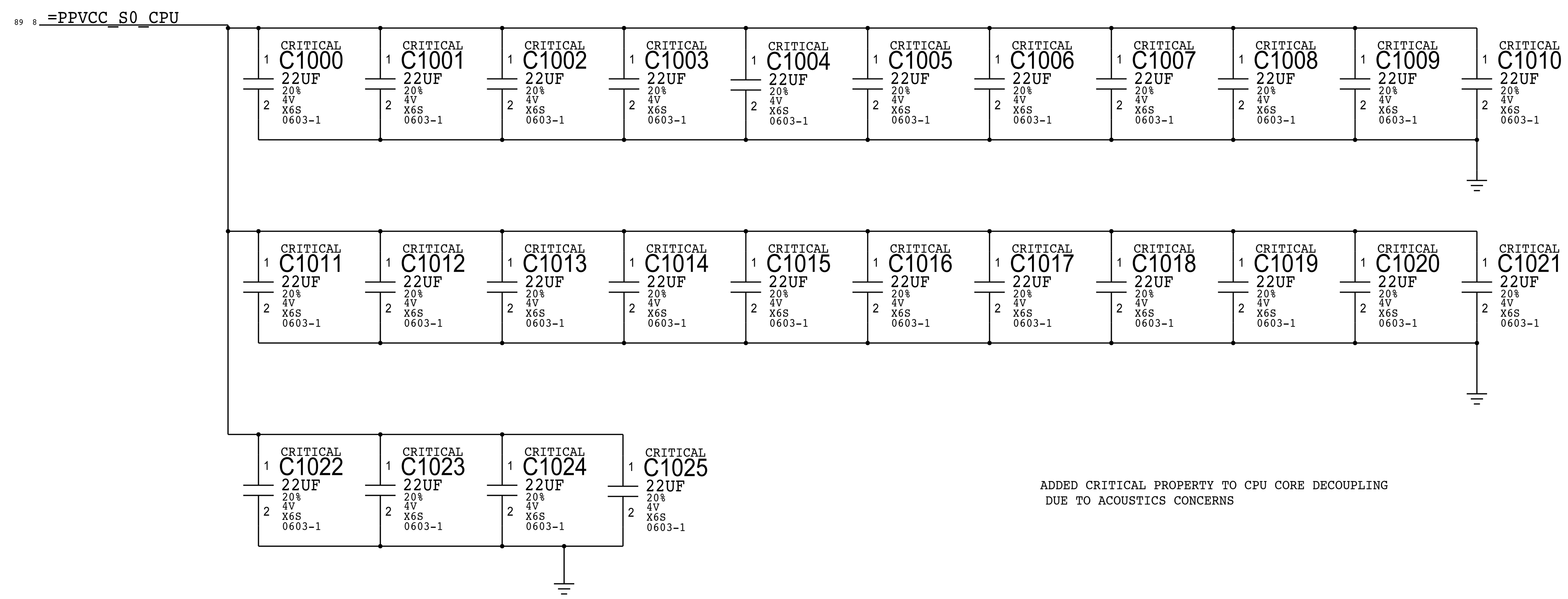
CPU VCORE DECOUPLING

Intel Recommendation:12x 22UF 0805 (top side cavity)
6x 22UF 0603 (top side cavity)
5x 22UF 0805 (top side outside cavity)

Apple Implementation:26x 22UF 0603

Layout Note: These caps should be placed symmetrically on Top and Bottom sides.
BULK CAPS ON CPU VREG PAGE 71



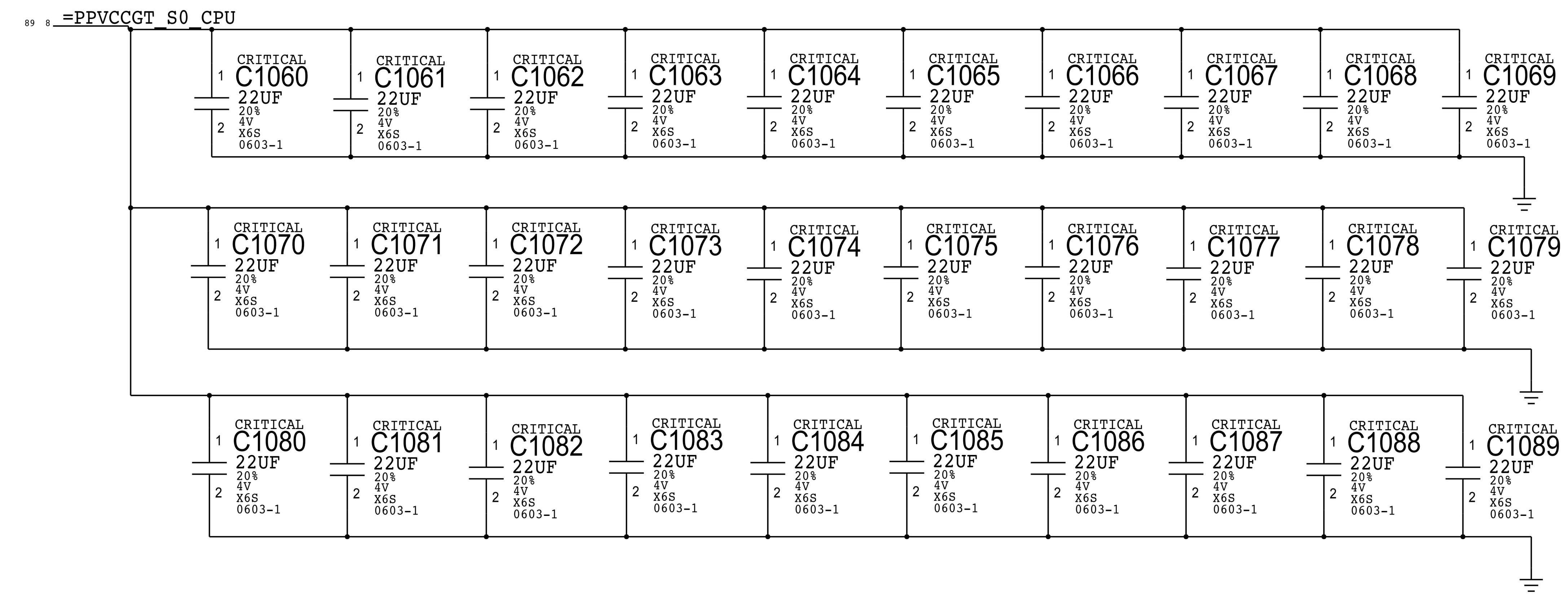
ADDED CRITICAL PROPERTY TO CPU CORE DECOUPLING
DUE TO ACOUSTICS CONCERNS

CPU GT DECOUPLING

Intel Recommendation:9x 47UF 0805 (top side cavity)
4x 47UF 0805 (top side outside cavity)

Apple Implementation: 30x 22uF 0603

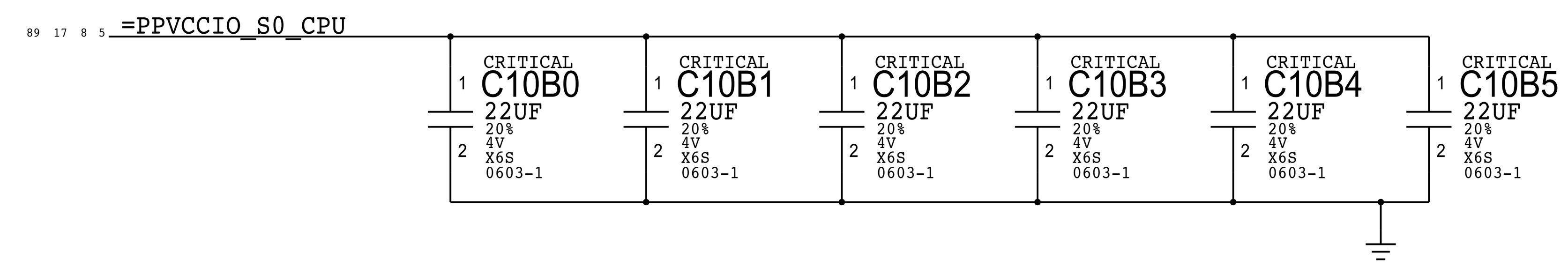
Layout Note: These caps should be placed symmetrically on Top and Bottom sides.



CPU VCCIO DECOUPLING

Intel Recommendation:5x 22UF 0603 (top side cavity)
1x 22UF 0805 (top side cavity)

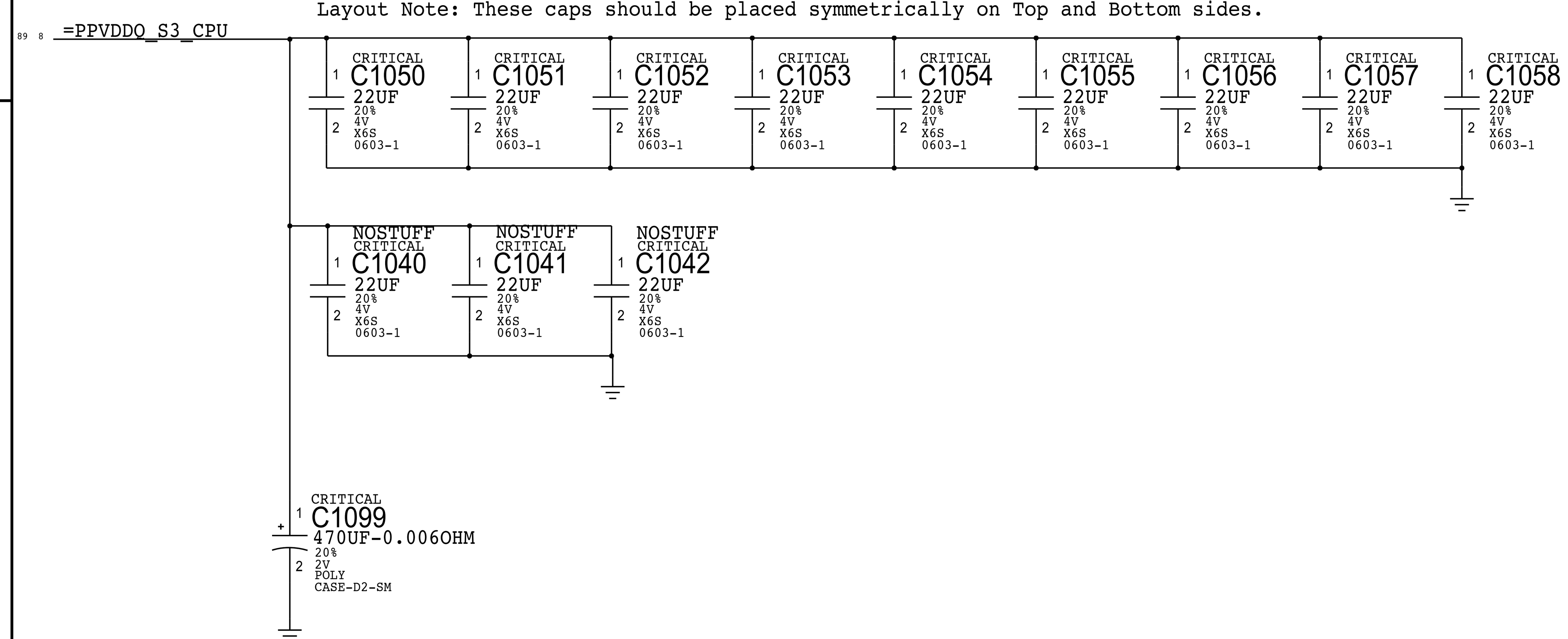
Apple Implementation:(following Intel recommendation w/ 0603)
Layout Note: These caps should be placed symmetrically on Top and Bottom sides.



Memory (CPU VCCDDR) DECOUPLING

Intel Recommendation:4x 22UF 0603 (top side outside cavity)

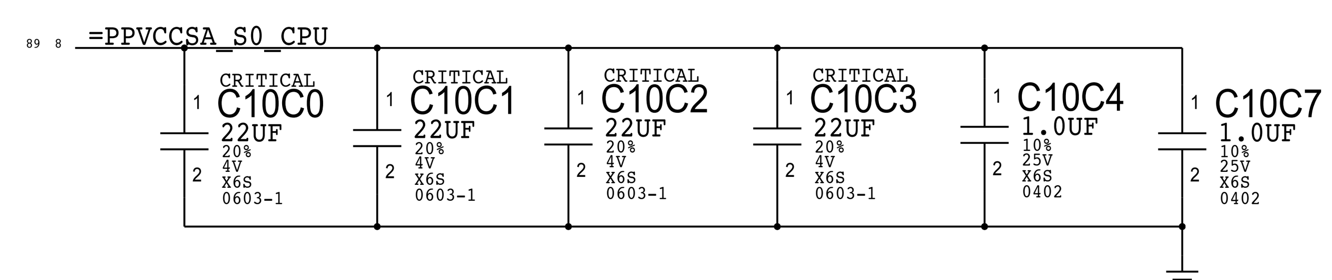
Apple Implementation:9x 22UF 0603 (J78 carry over)
Layout Note: These caps should be placed symmetrically on Top and Bottom sides.

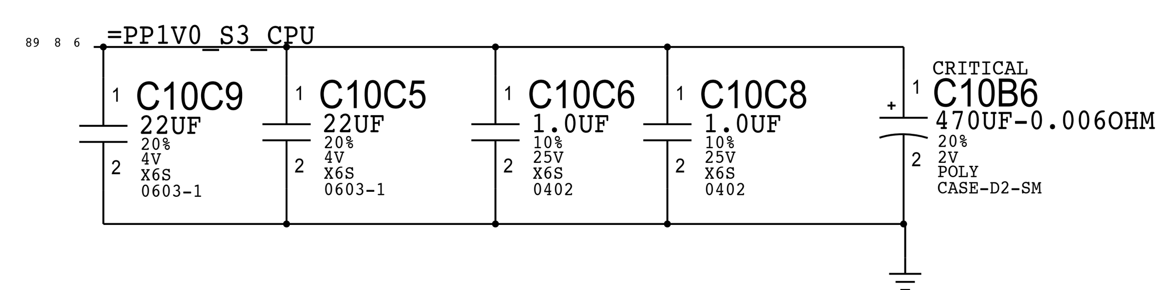


CPU VCCSA / VCCST+VCCPLL DECOUPLING

Intel Recommendation:2x 22UF 0603 near top side cavity

Apple Implementation: VCCST/VCCPLL: 1X 22UF 0603/2X 1UF 0402
Layout Note: These caps should be placed on top side cavity.





SYNC_MASTER=J78_MLB	SYNC_DATE=06/30/2014
PAGE TITLE: CPU & CHIPSET: CPU Decoupling	
Apple Inc.	DRAWING NUMBER 051-00673, SIZE D
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SYNC_MASTER=J95_DEBORAH SYNC_DATE=10/20/2014

PAGE TITLE

CPU & CHIPSET: Clocks/HDA/JTAG

Apple Inc.

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SYNC_MASTER=J95_DEBORAH SYNC_DATE=10/20/2014

PAGE TITLE

CPU & CHIPSET: PCH RTC/DMI/PM/CPU_Misc

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SYNC_MASTER=J95_DEBORAH SYNC_DATE=10/20/2014

PAGE TITLE

CPU & CHIPSET: PCH PCI-E/USB

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SYNC_MASTER=J95_DEBORAH SYNC_DATE=10/20/2014

PAGE TITLE

CPU & CHIPSET: PCH GPIO/Misc

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NEED TO ADD 47UF BULK CAPS TO 1.0V AND 3.3V RAIL

Current data from EDS (doc #546717, Rev 0.7).

SYNC_MASTER=J95_DEBORAH	SYNC_DATE=10/20/2014
PAGE TITLE	CPU & CHIPSET: PCH Power/Decoupling
DRAWING NUMBER	051-00673
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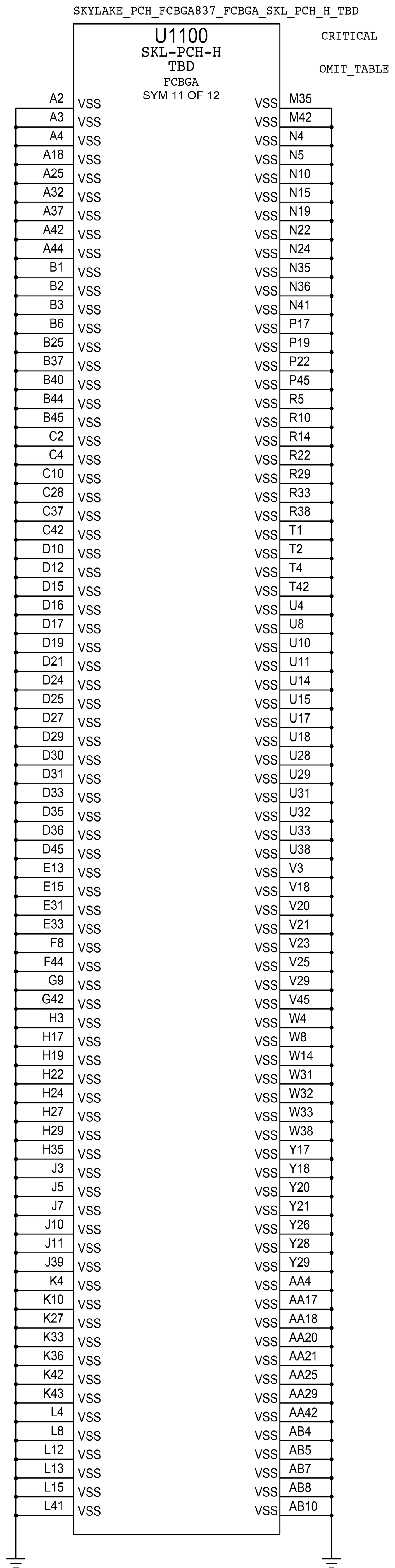

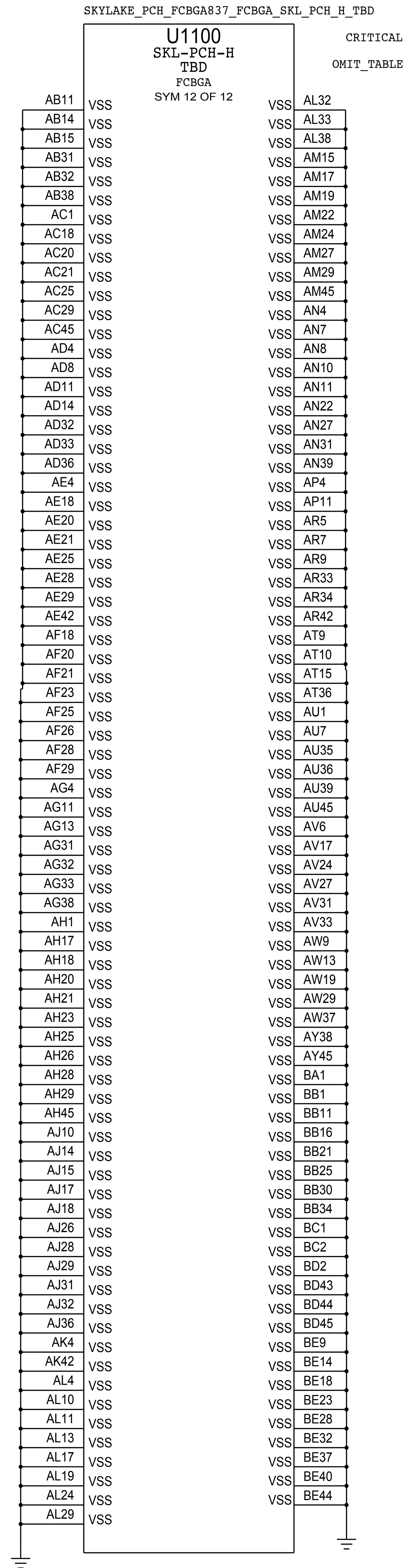

SYNC_MASTER=J95_DEBORAH SYNC_DATE=10/20/2014

PAGE TITLE

CPU & CHIPSET: PCH Grounds

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SYNC_MASTER=J95_ANDREW SYNC_DATE=01/24/2015

PAGE TITLE

CPU & CHIPSET: XDP

DRAWING NUMBER 051-00673 SIZE D

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REVISION 0.24.0

BRANCH

SYNC_MASTER=BRANCH_SYEDKAR SYNC_DATE=09/10/2014

PAGE TITLE

CPU & CHIPSET: Chipset Support

Apple Inc.

DRAWING NUMBER 051-00673 SIZE D

REVISION 0.24.0

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SYNC_MASTER=J16_IG SYNC_DATE=04/29/2013

PAGE TITLE

CPU & CHIPSET: Project Chipset Support

DRAWING NUMBER 051-00673 SIZE D

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REVISION 0.24.0

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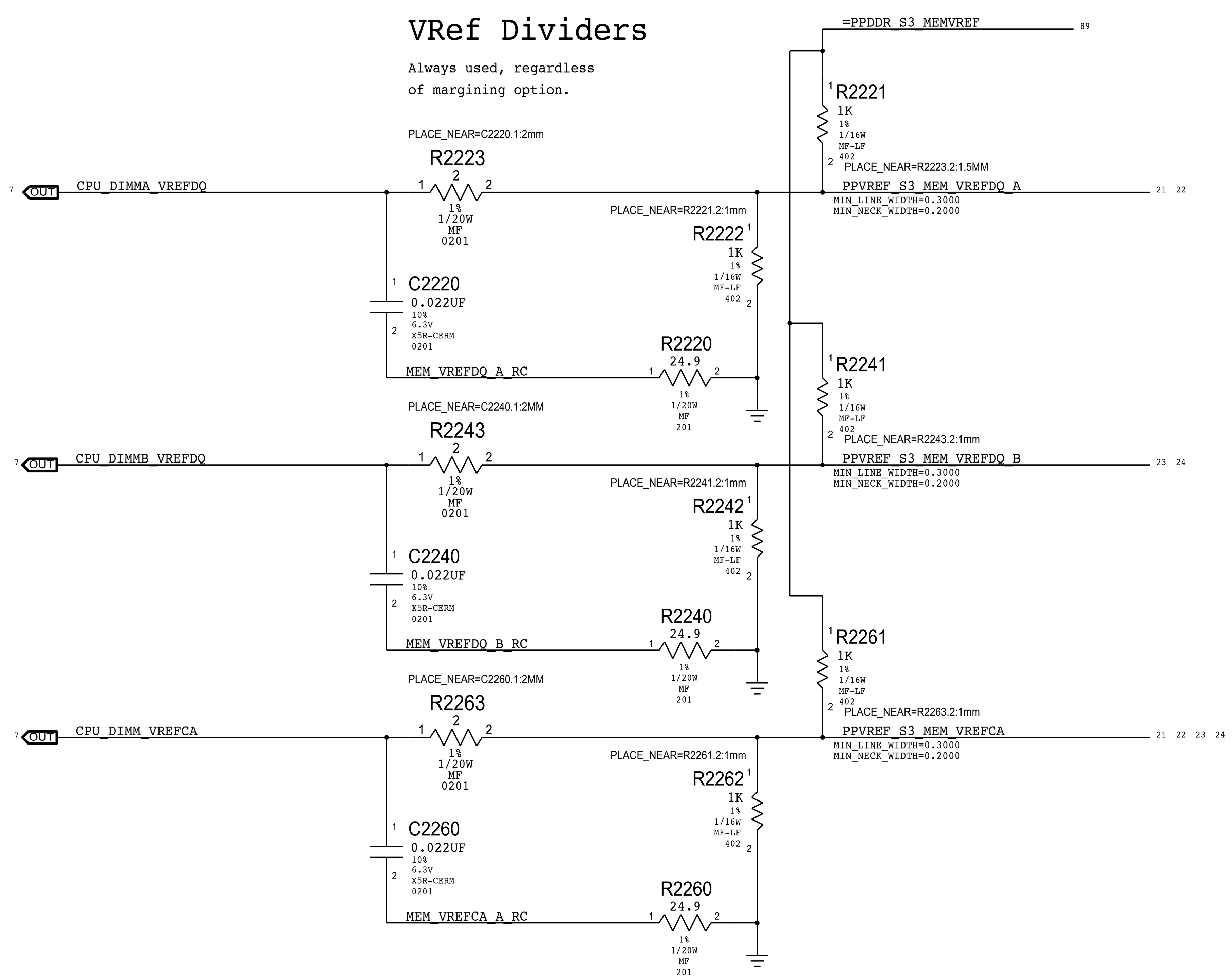

SYNC_MASTER=J78_DAVID SYNC_DATE=03/10/2014

PAGE TITLE

DRAM: VREF/VTT EN

Apple Inc.

DRAWING NUMBER	SIZE
051-00673	D

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Page Notes

Power aliases required by this page:

- =PP1V5_S0_MEM_A
- =PPVDDQ_S3_MEM_A
- =PPDDRVTT_S0_MEM_A
- =PPSPD_S0_MEM_A (2.5 - 3.3V)

Signal aliases required by this page:

- =I2C_SODIMMA_SCL
- =I2C_SODIMMA_SDA

BOM options provided by this page:

(NONE)

SYNC_MASTER=J78_NAT

SYNC_DATE=10/30/2013

PAGE TITLE

DRAM: SO-DIMM Connector A Slot0

Apple Inc.

DRAWING NUMBER 051-00673

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Page Notes

Power aliases required by this page:

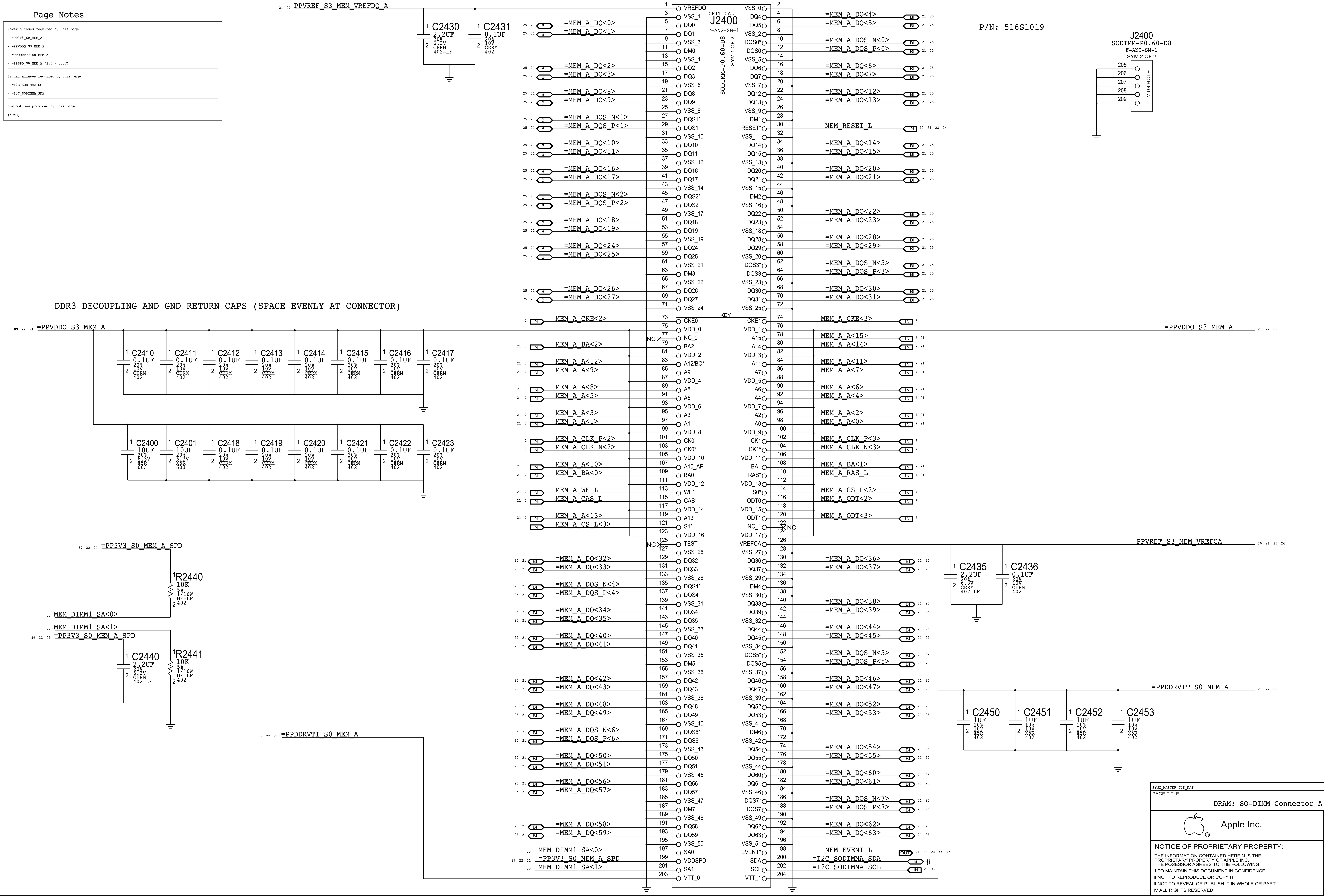
- =PP1V5_S0_MEM_A
- =PPVDDQ_S3_MEM_A
- =PPDDRVTT_S0_MEM_A
- =PPSPD_S0_MEM_A (2.5 - 3.3V)

Signal aliases required by this page:

- =I2C_SODIMMA_SCL
- =I2C_SODIMMA_SDA

BOM options provided by this page:

(NONE)



SYNC_MASTER=J78_MAT SYNC_DATE=10/30/2013

PAGE TITLE

DRAM: SO-DIMM Connector A Slot1

DRAWING NUMBER 051-00673 SIZE D

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REVISION 0.24.0

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8 7 6 5 4 3 2 1

Page Notes

Power aliases required by this page:

- =PP1V5_S0_MEM_B
- =PPVDDQ_S3_MEM_B
- =PPDDRVTT_S0_MEM_B
- =PPSPD_S0_MEM_B (2.5 - 3.3V)

Signal aliases required by this page:

- =I2C_SODIMMA_SCL
- =I2C_SODIMMA_SDA

BOM options provided by this page:

(NONE)

SYNC_MASTER=J78_NAT SYNC_DATE=10/30/2013

PAGE TITLE

DRAM: SO-DIMM CONNECTOR B SLOT0

Apple Inc.

DRAWING NUMBER	SIZE
051-00673	D

REVISION: 0.24.0

BRANCH

PAGE: 25 OF 121

SHEET: 23 OF 93

Page Notes
Power aliases required by this page:
DDR3 DECOUPLING AND GND RETURN CAPS (SPACE EVENLY AT CONNECTOR)
P/N: 516S1019
PPVREF_S3_MEM_VREFDQ_B
VOLTAGE=0.675V
C2630 2.2UF 20% 6.3V CERM 402-LF
C2631 0.1UF 20% 10V CERM 402
CRITICAL
J2600
F-ANG-SM-1
SODIMM-P0.60-D8
SYM 1 OF 2
J2600
SODIMM-P0.60-D8
F-ANG-SM-1
SYM 2 OF 2
MTG HOLE
=PPVDDQ_S3_MEM_B
C2610 C2611 C2612 C2613 C2614 C2615 C2616 C2617 0.1UF 20% 10V CERM 402
C2600 C2601 10UF 20% 6.3V X5R 603
C2618 C2619 C2620 C2621 C2622 C2623 0.1UF 20% 10V CERM 402
=PP3V3_S0_MEM_B_SPD
R2641 10K 5% 1/16W MF-LF 402
R2640 10K 5% 1/16W MF-LF 402
MEM_DIMM3_SA<1>
MEM_DIMM3_SA<0>
C2640 2.2UF 20% 6.3V CERM 402-LF
=PPDDRVTT_S0_MEM_B
MEM_RESET_L
MEM_B_CKE<2>
MEM_B_CKE<3>
PPVREF_S3_MEM_VREFCA
C2635 2.2UF 20% 6.3V CERM 402-LF
C2636 0.1UF 20% 10V CERM 402
C2650 C2651 C2652 C2653 1UF 10% 10V X5R 402
MEM_EVENT_L
=I2C_SODIMMB_SDA
=I2C_SODIMMB_SCL
SYNC_MASTER=J78_NAT
SYNC_DATE=10/30/2013
PAGE TITLE
DRAM: SO-DIMM CONNECTOR B SLOT1
Apple Inc.
DRAWING NUMBER
051-00673
SIZE
D
REVISION
0.24.0
BRANCH
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26 OF 121
SHEET
24 OF 93
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IV ALL RIGHTS RESERVED

THERE ARE NO PIN SWAPS

Page	Net	Property	Alias	Pages
7	MEM_A_DQS_N<0>	MAKE_BASE=TRUE	=MEM_A_DQS_N<0>	21 22
7	MEM_A_DQS_P<0>	MAKE_BASE=TRUE	=MEM_A_DQS_P<0>	21 22
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7	MEM_A_DQS_P<4>	MAKE_BASE=TRUE	=MEM_A_DQS_P<4>	21 22
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Page	Net	Property	Alias	Pages
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SYNC_MASTER=J16_IG SYNC_DATE=04/29/2013

PAGE TITLE

DRAM: ALIASES AND BITSWAPS

Apple Inc.

DRAWING NUMBER 051-00673 SIZE D

REVISION 0.24.0

BRANCH

SYNC_MASTER=BRANCH_JERRYCHOW SYNC_DATE=11/06/2014

PAGE TITLE

TBT/DP: Host (1 of 2)

DRAWING NUMBER 051-00673 SIZE D

Apple Inc.

REVISION 0.24.0

BRANCH

SYNC_MASTER=BRANCH_JERRYCHOW | SYNC_DATE=09/10/2014

PAGE TITLE: TBT/DP: Host (2 of 2)

Apple Inc.

DRAWING NUMBER: 051-00673 | SIZE: D

REVISION: 0.24.0

BRANCH

PAGE: 29 OF 121

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EDP current / power consumption figures copied from R68 schematic (Rev 2, dated October 28, 2012, not available on IBL).

Page Notes

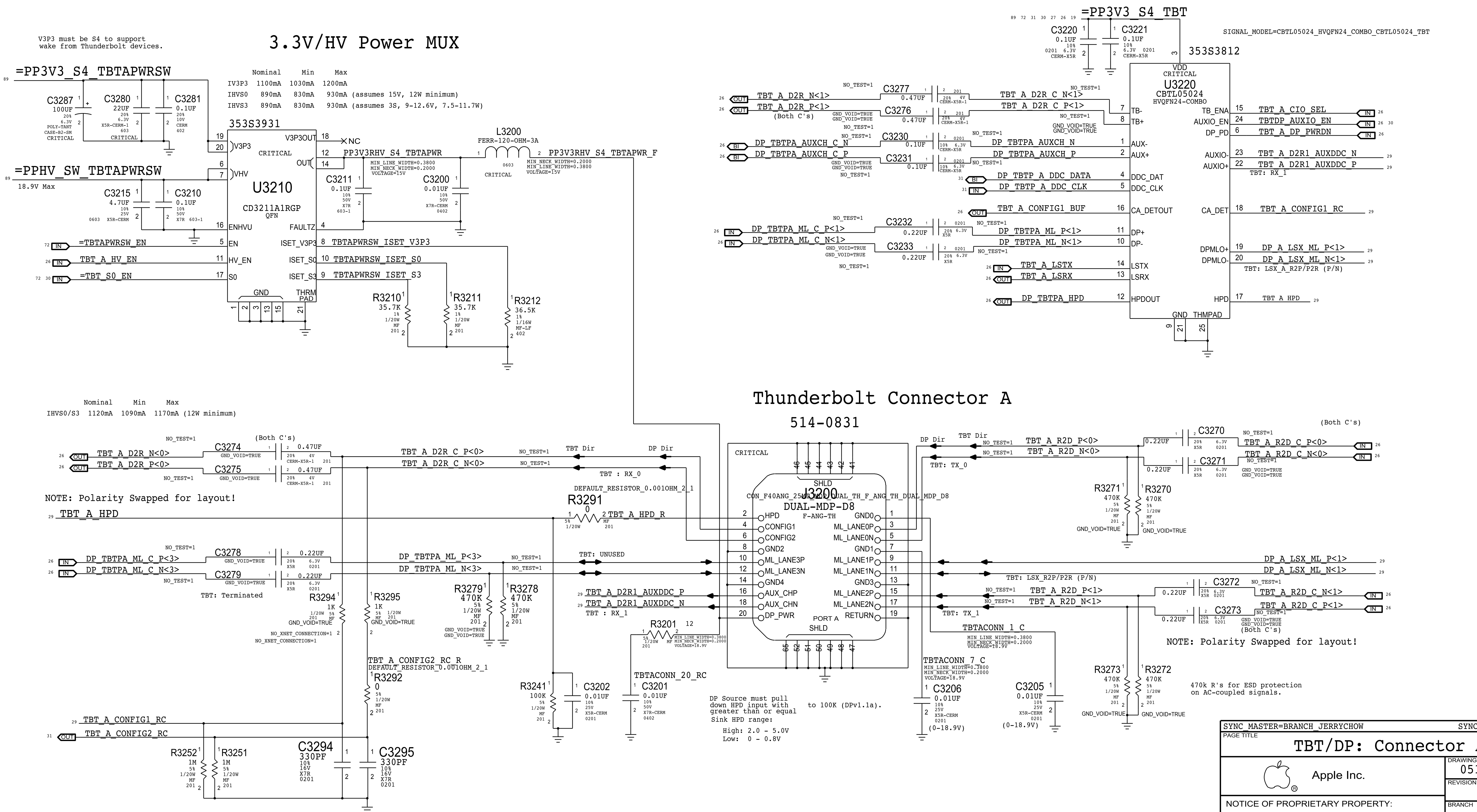
Power aliases required by this page:
- =PPVIN_SW_TBTBST (8-13V Boost Input)
- =PP15V_TBT_REG (15V Boost Output)
- =PP3V3_TBT_P3V3TBTFET (3.3V FET Input)
- =PP3V3_TBT_FET (3.3V FET Output)
- =PP3V3_S0_TBTPWRCTL
- =PP1V05_TBT_P1V05TBTFET (1.05V FET Input)
- =PP1V05_TBT_FET (1.05V FET Output)

Signal aliases required by this page:
- =TBT_CLKREQ_L
- =TBT_RESET_L

BOM options provided by this page:
TBTBST:Y - Stuffs 15V boost circuitry.

3.3V/HV Power MUX

Thunderbolt Connector A



SYNC_MASTER=BRANCH_JERRYCHOW	SYNC_DATE=09/10/2014
PAGE TITLE	TBT/DP: Connector A
DRAWING NUMBER	051-00673
SIZE	D
Apple Inc.	
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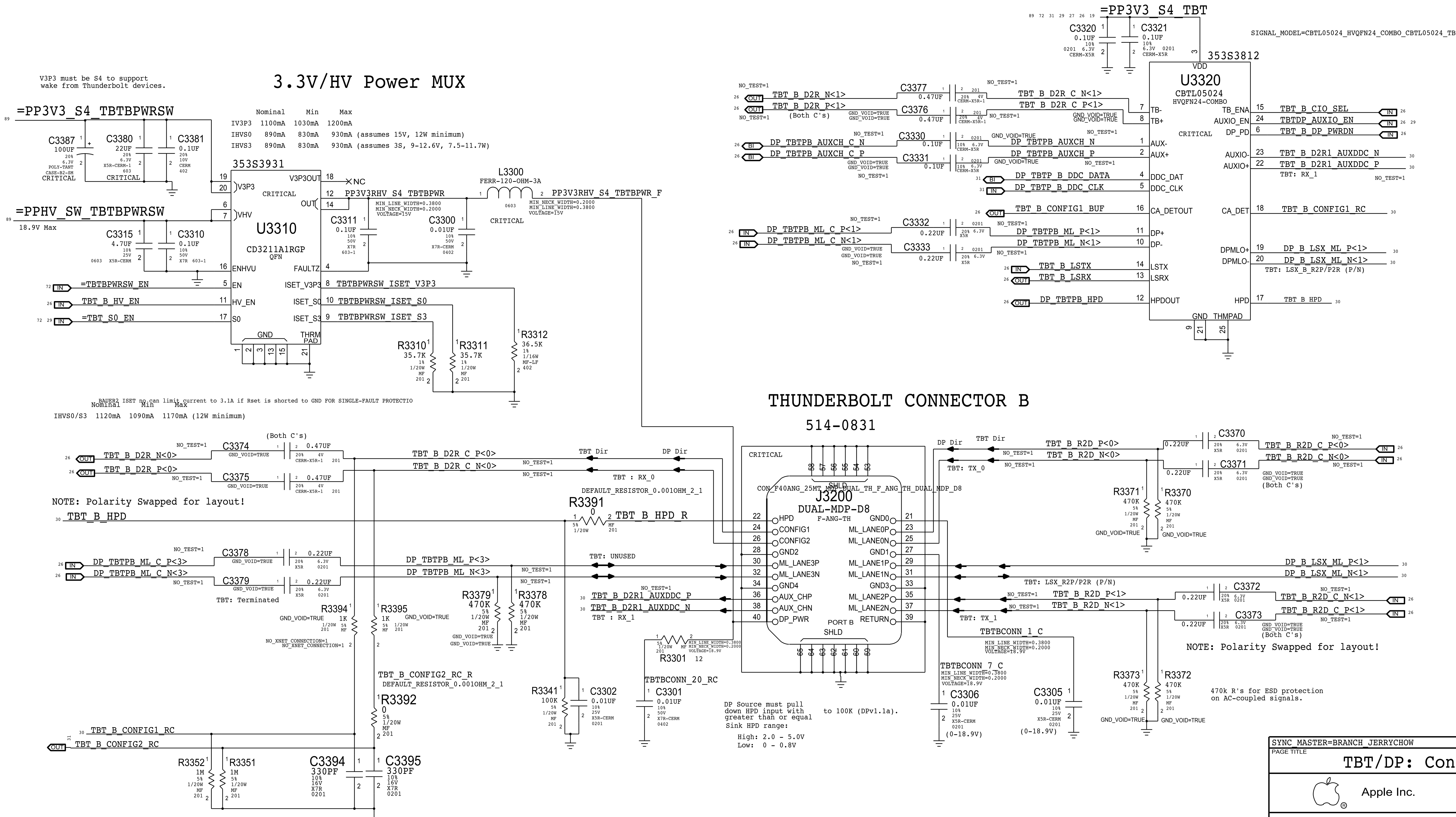
8 7 6 5 4 3 2 1

D

C

B

A



SYNC_MASTER=BRANCH_JERRYCHOW	SYNC_DATE=09/10/2014
PAGE TITLE	TBT/DP: Connector B
Apple Inc.	DRAWING NUMBER 051-00673 SIZE D
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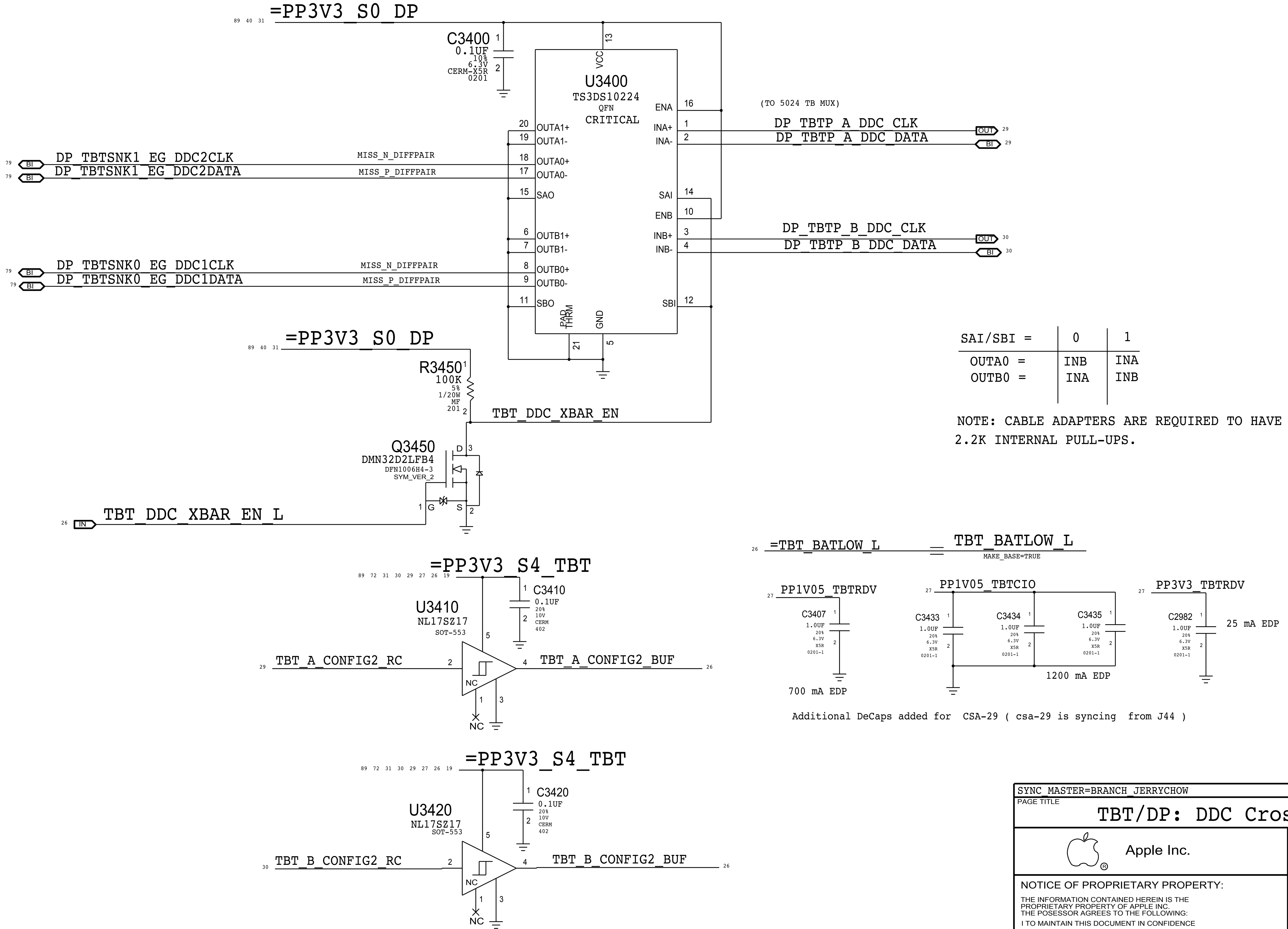
	SHEET 30 OF 93

8 7 6 5 4 3 2 1

Dual-Port Host DDC Crossbar



SYNC_MASTER=BRANCH_JERRYCHOW SYNC_DATE=11/05/2014

PAGE TITLE: TBT/DP: DDC Crossbar

DRAWING NUMBER: 051-00673 SIZE: D

Apple Inc.

REVISION: 0.24.0

BRANCH

AIRPORT
BLUETOOTH

AP & BT Load Switch

SWITCH	TPS22924B
CHANNEL	N-TYPE
RDS(ON)	18.4 MOHM @3.3V
LOADING	2 A (EDP)

SYNC_MASTER=J16_IG SYNC_DATE=04/29/2013

PAGE TITLE: WIRELESS: Airport/Bluetooth

Apple Inc.

DRAWING NUMBER: 051-00673 SIZE: D

REVISION: 0.24.0

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SYNC_MASTER=J16_IG SYNC_DATE=04/29/2013

PAGE TITLE: SDD/HDD:SATA/SSD Connectors

Apple Inc.

DRAWING NUMBER: 051-00673 SIZE: D

REVISION: 0.24.0

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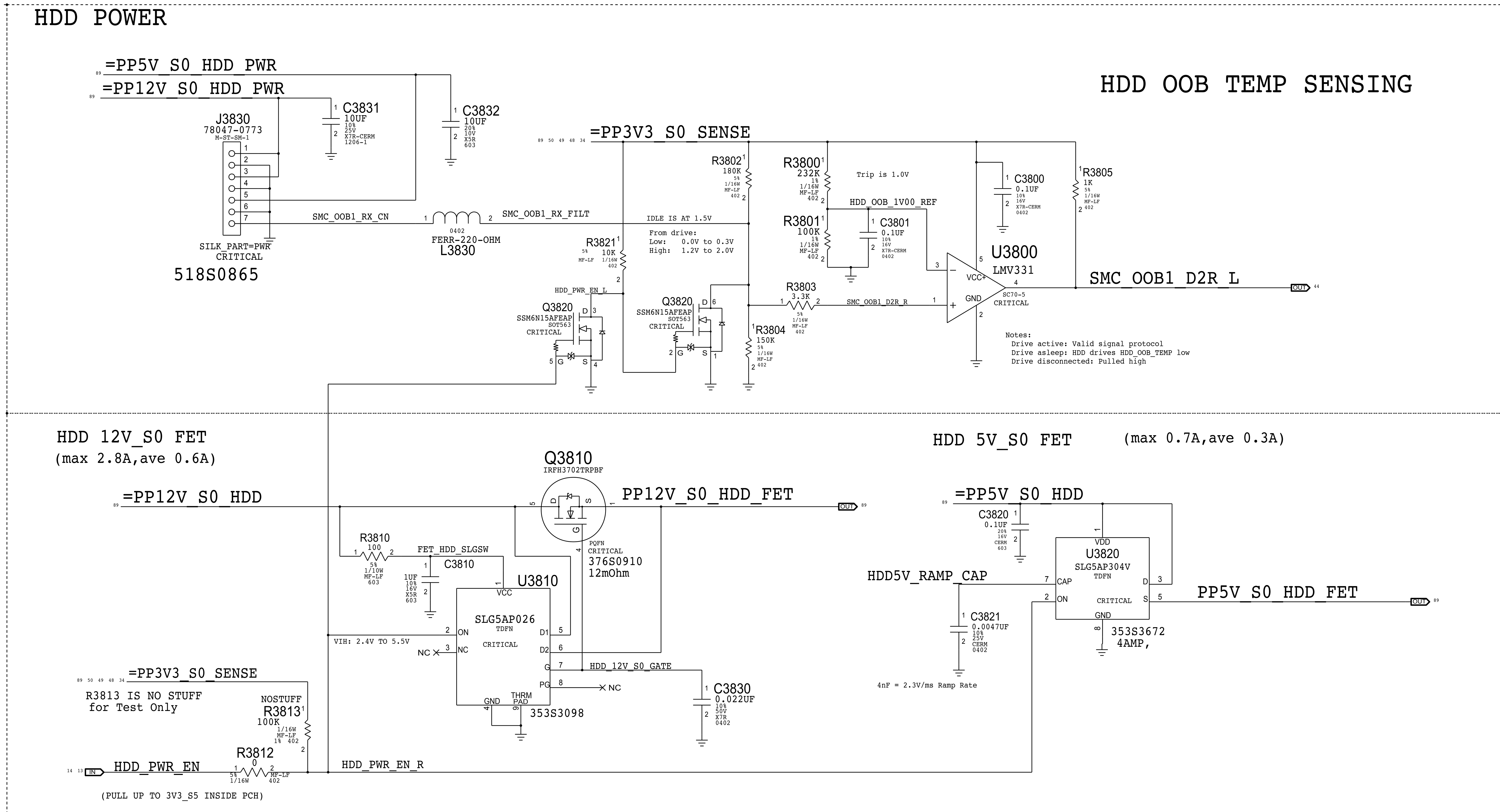
NOTICE OF PROPRIETARY PROPERTY:



REMOVE R3813 AND SHORT R3812 AFTER HDD_PWR_EN WORKS

SYNC_MASTER=J78_KENNY SYNC_DATE=03/10/2014

PAGE TITLE

HDD: SSD Temp Sense

DRAWING NUMBER 051-00673 SIZE D

Apple Inc.

REVISION 0.24.0

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SYNC_MASTER=J16_IG SYNC_DATE=04/29/2013

PAGE TITLE: ETHERNET: PHY (CAESAR IV)

Apple Inc.

DRAWING NUMBER: 051-00673 SIZE: D

REVISION: 0.24.0

BRANCH

SYNC_MASTER=J16_IG SYNC_DATE=12/07/2012

PAGE TITLE

ETHERNET: Support & Connector

Apple Inc.

DRAWING NUMBER 051-00673 SIZE D

REVISION 0.24.0

BRANCH

SYNC_MASTER=J16_IG SYNC_DATE=04/29/2013

PAGE TITLE

SD CARD: Connector

Apple Inc.

DRAWING NUMBER 051-00673 SIZE D

REVISION 0.24.0

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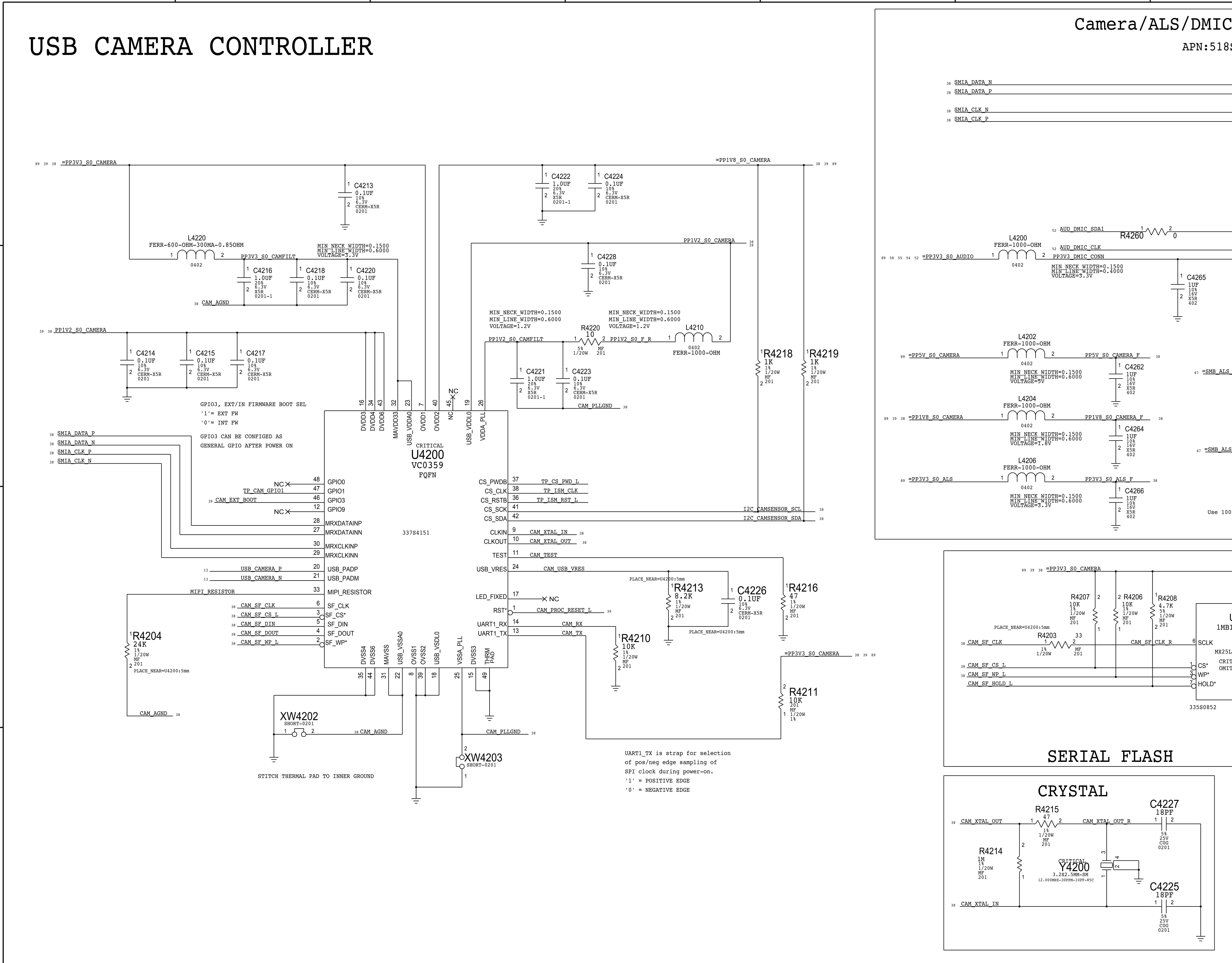

SYNC_MASTER=J78_NAT SYNC_DATE=11/05/2013

PAGE TITLE: CAMERA: Controller

DRAWING NUMBER: 051-00673 SIZE: D

Apple Inc.

REVISION: 0.24.0

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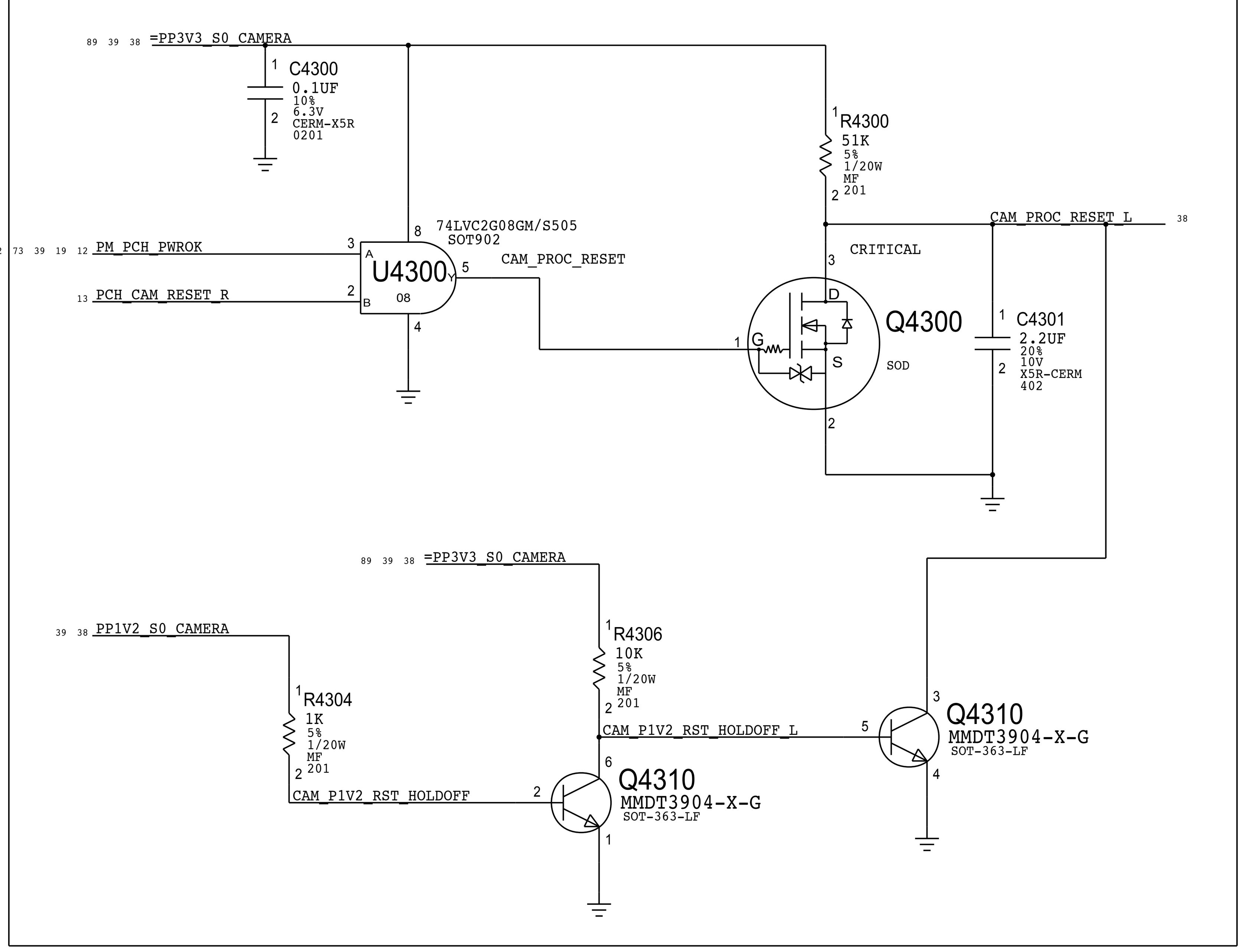
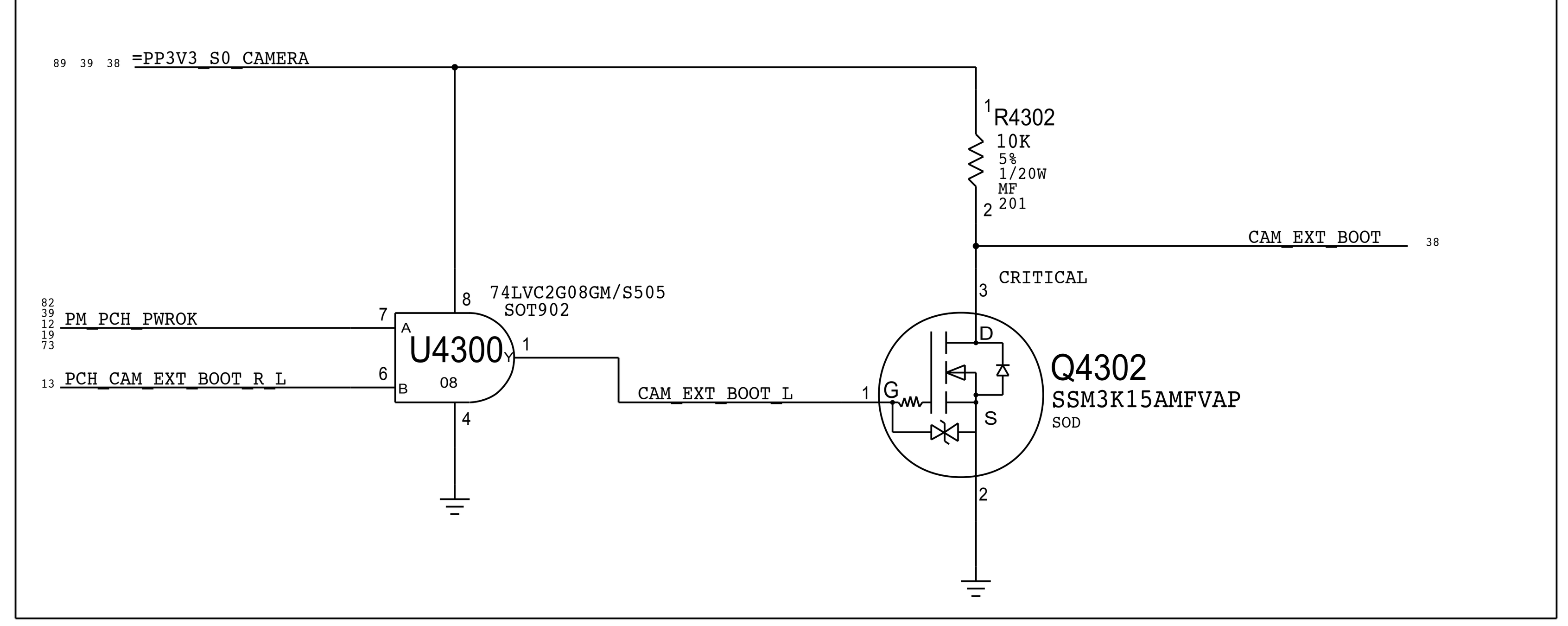
SHEET: 38 OF 93

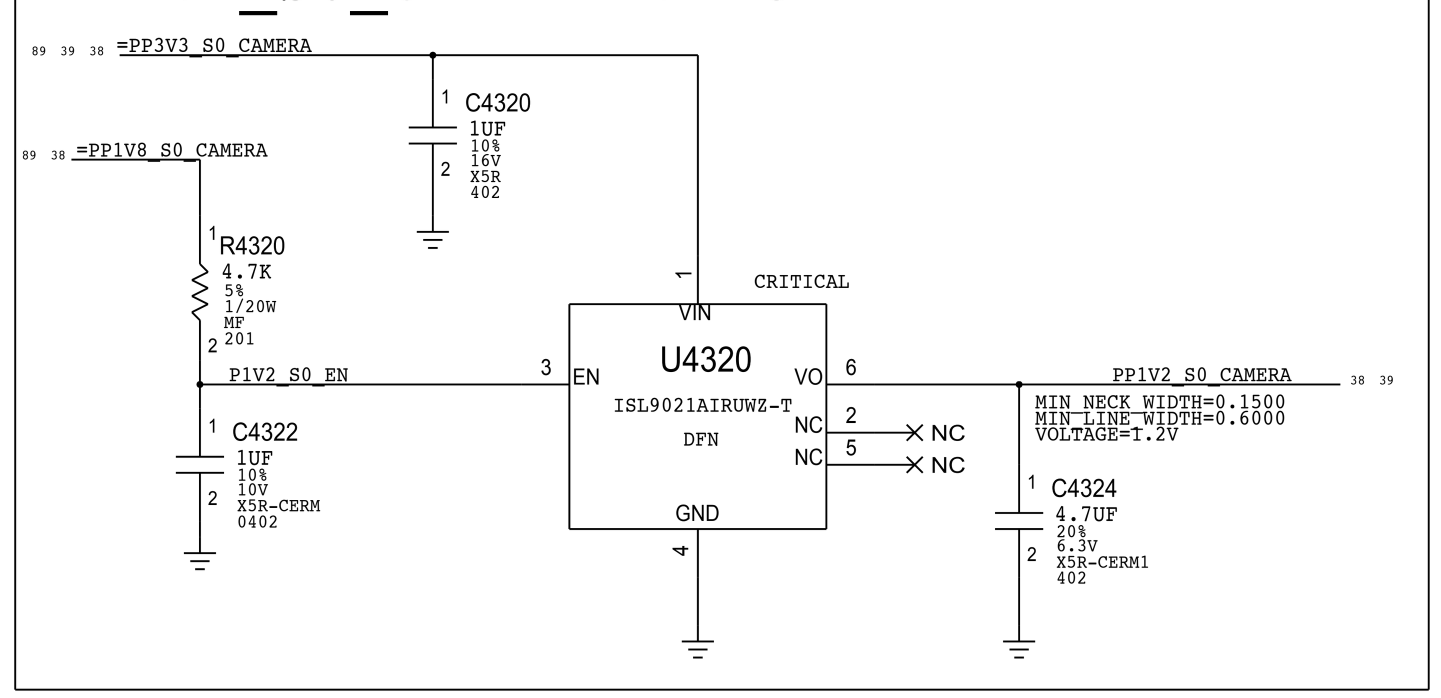
Camera Processor Reset



Camera Processor ExtBoot Cntl



PP1V2_S0_CAMERA VREG



SYNC_MASTER=J78_NAT SYNC_DATE=11/05/2013

PAGE TITLE
CAMERA: Controller Support

DRAWING NUMBER 051-00673 SIZE D

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REVISION 0.24.0

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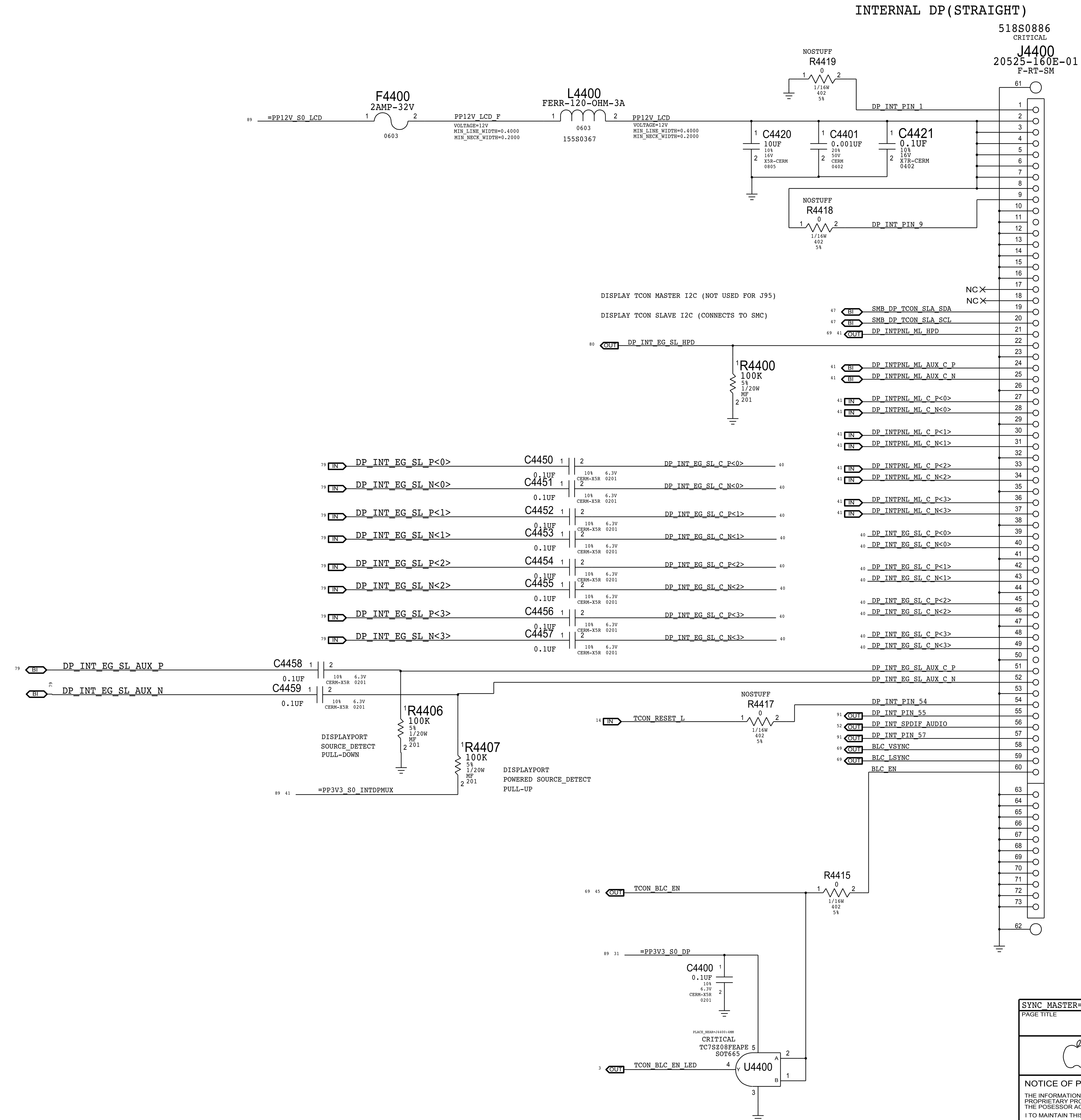

SYNC_MASTER=J95_ANDREW SYNC_DATE=1/27/2015

PAGE TITLE

DISPLAY: Support

Apple Inc.

DRAWING NUMBER 051-00673 SIZE D

REVISION 0.24.0

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SYNC_MASTER=J78_KENNY SYNC_DATE=10/09/2013

PAGE TITLE

DISPLAY: MUXing

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SYNC_MASTER=J78_DAVID SYNC_DATE=10/21/2013

PAGE TITLE

USB-A: EXTERNAL USB PORTS A & B

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BRANCH	
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SYNC_MASTER=J16_IG SYNC_DATE=04/29/2013

PAGE TITLE

USB-A: EXTERNAL USB PORTS C & D

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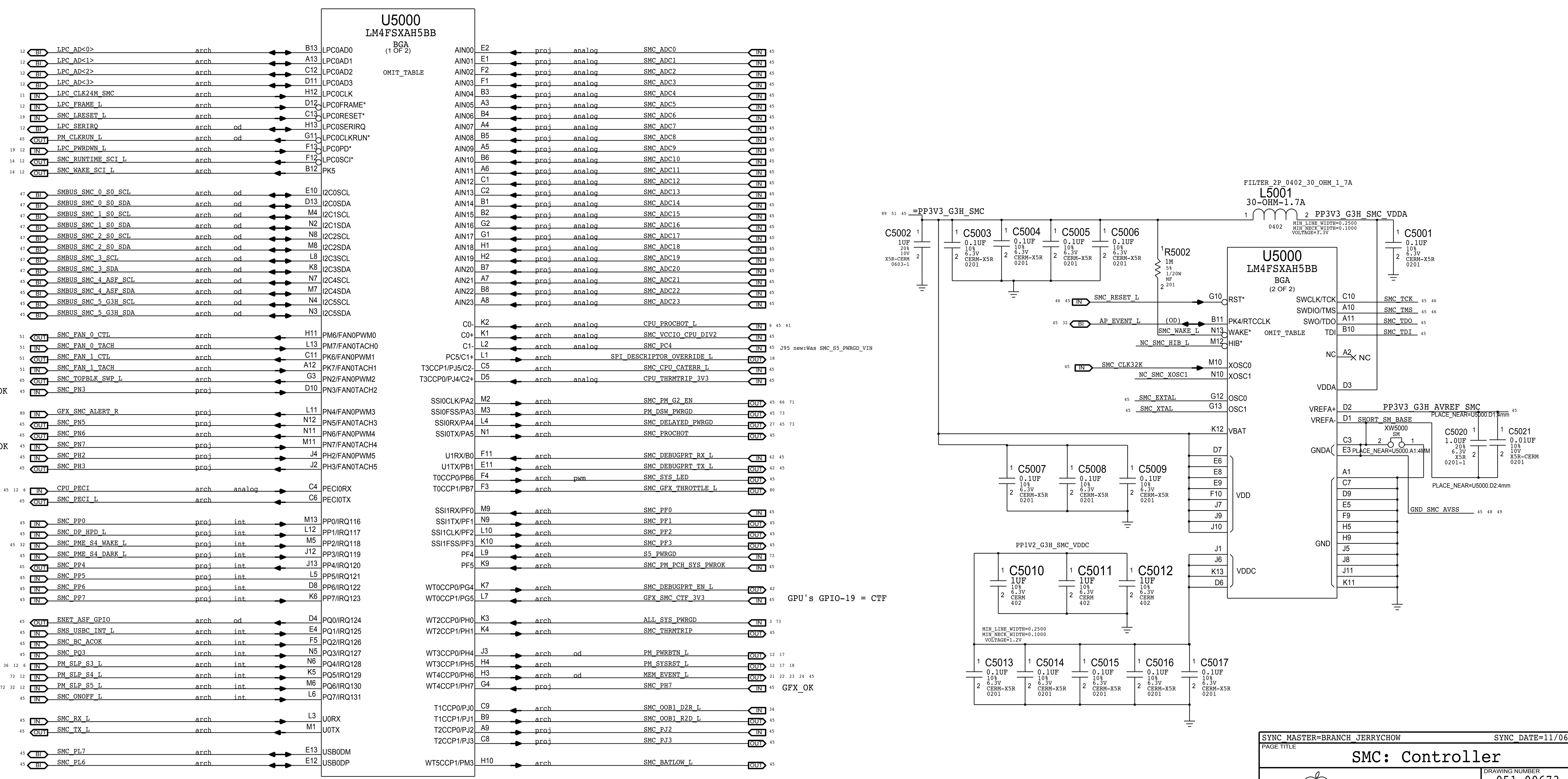
NOTE: Unused pins have "SMC_Pxx" names. Unused pins designed as outputs can be left floating, those designated as inputs require pull-ups.



SYNC_MASTER=BRANCH_JERRYCHOW SYNC_DATE=11/06/2014

PAGE TITLE

SMC: Controller

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SMC Supervisor and AVREF Supply
ADC Channel Aliases
Project-specific Aliases
Unused Project-specific
SMC 32KHz Clock
SMC Crystal
SMC Controlled RTC Reset
Power Button
PECI Support
Comparator VRef
AC/DC Burst Mode Enable
Platform Thermal Control
PROCHOT Support
Arch Pull Up/Down
Serial/JTAG Interface Pull-ups
Note: IPU are pulled to VIN rail
SYNC_MASTER=BRANCH_JERRYCHOW
SYNC_DATE=11/06/2014
PAGE TITLE
SMC: Controller Support
DRAWING NUMBER
051-00673
SIZE
D
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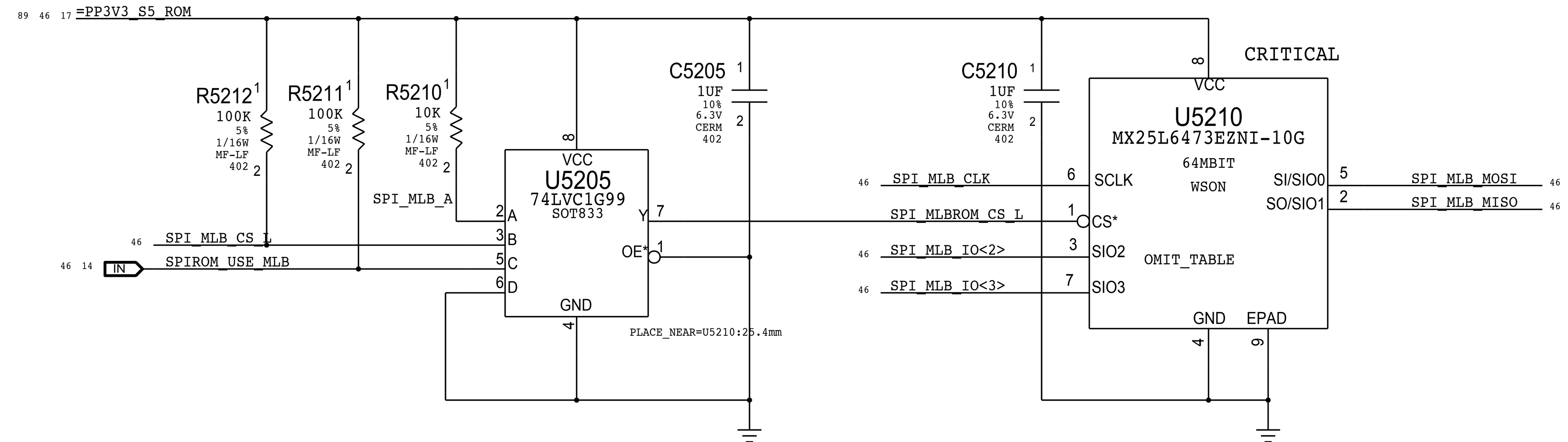
SHEET
45 OF 93

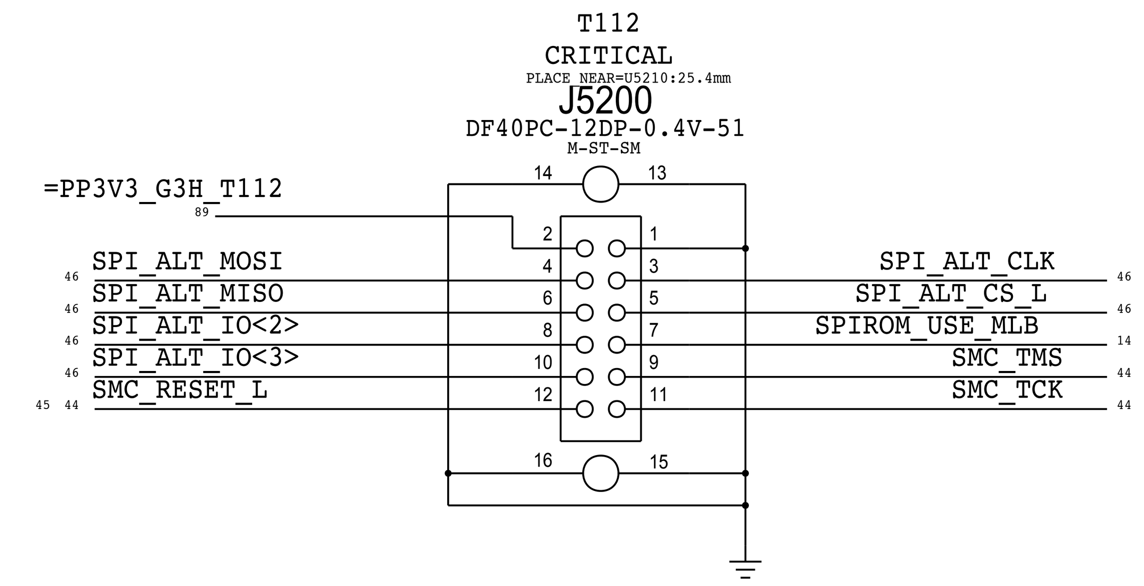
SPI BootROM

QUAD_IO MODE (MODE 0 & 3) SUPPORTED

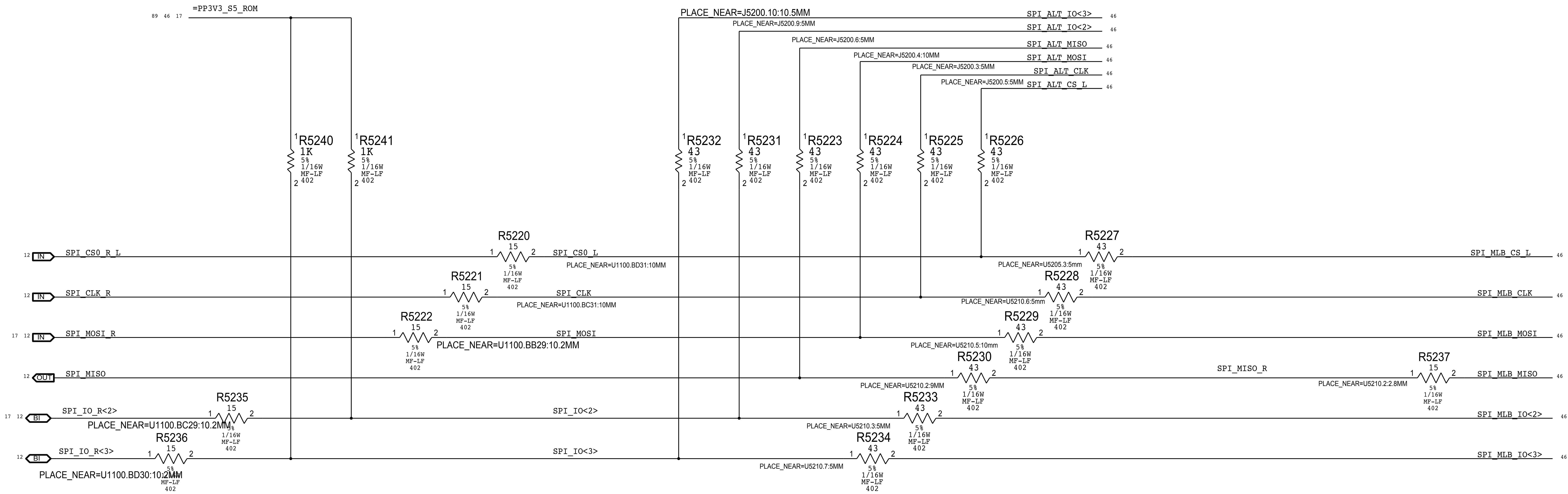
SPI FREQ: 50 MHZ FOR PCH



T112 Connector



SPI Series Termination



SYNC_MASTER=J17_TONY SYNC_DATE=03/13/2013

PAGE TITLE: CPU & CHIPSET: SPI and Debug Connector

Apple Inc.

DRAWING NUMBER: 051-00673 SIZE: D

REVISION: 0.24.0

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SYNC_MASTER=BRANCH_JERRYCHOW SYNC_DATE=08/26/2014

PAGE TITLE

SMC: SMBus Connections

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SYNC_MASTER=BRANCH_JERRYCHOW SYNC_DATE=09/10/2014

PAGE TITLE

SENSORS: I and V Sense

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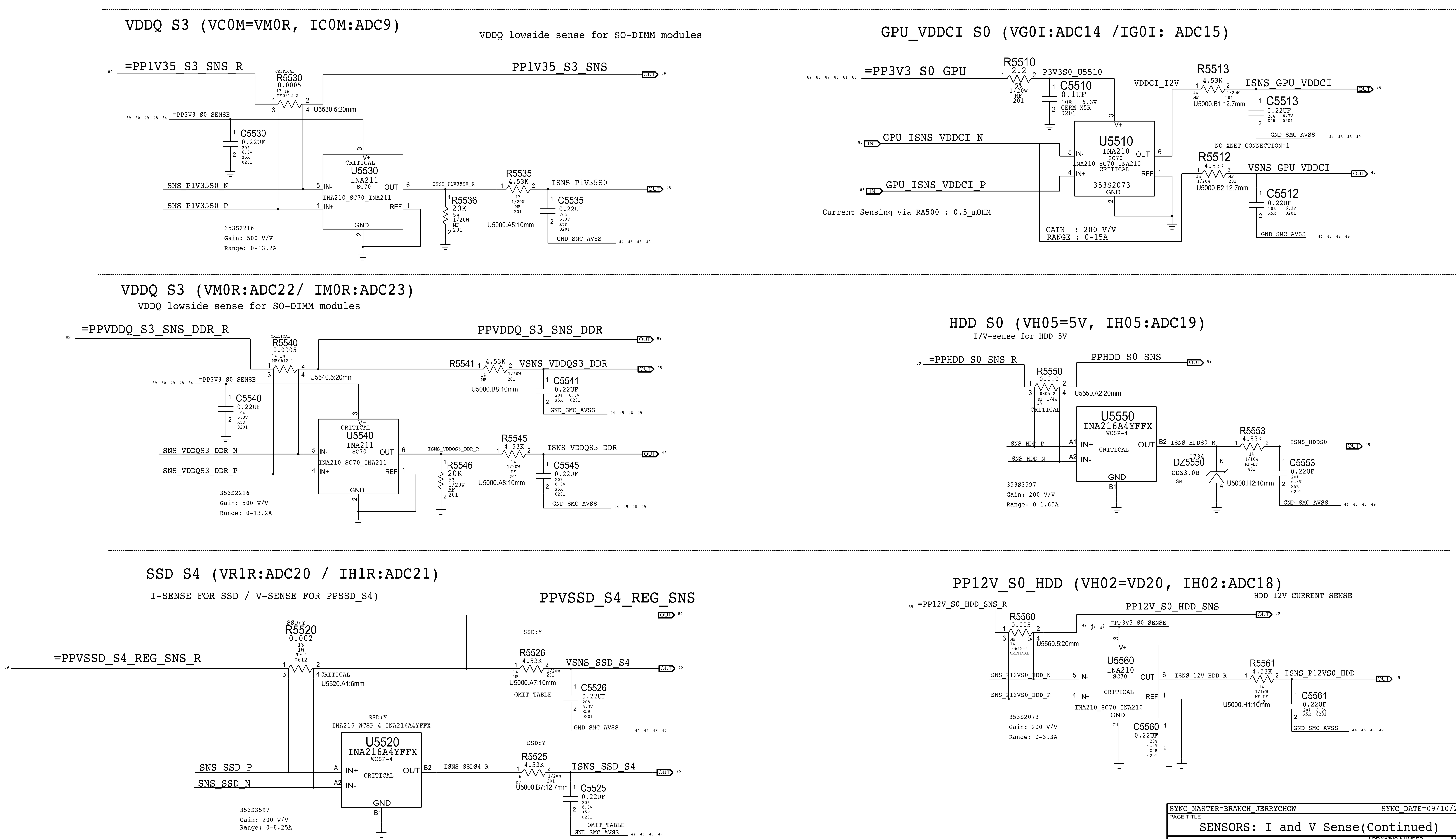

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
132S0304	2	CAP,0.22UF,201	C5525,C5526	SSD:Y
117S0201	2	RES,0 OHM,201	C5525,C5526	SSD:N

SYNC_MASTER=BRANCH_JERRYCHOW SYNC_DATE=09/10/2014

PAGE TITLE

SENSORS: I and V Sense(Continued)

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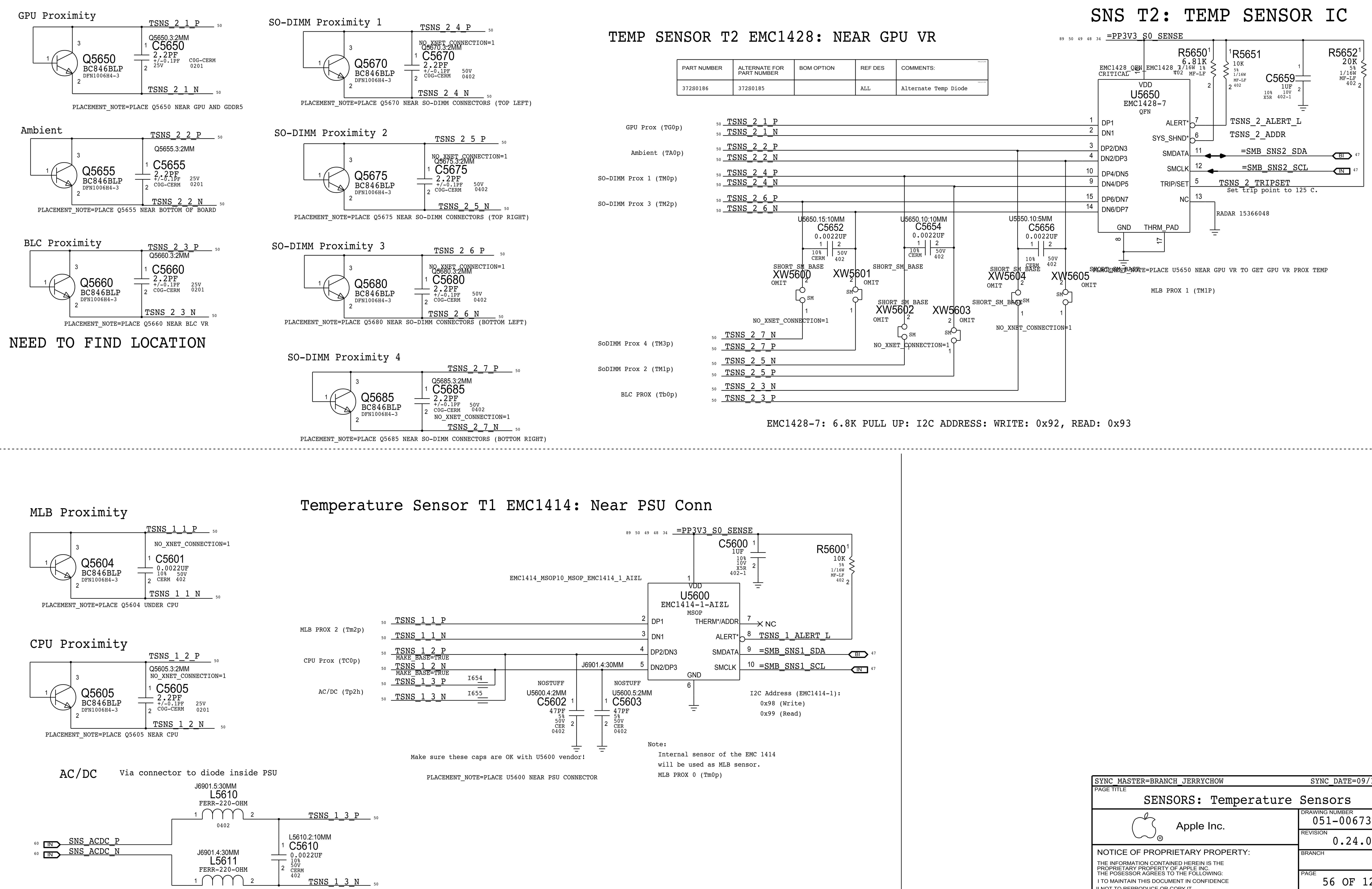
8 7 6 5 4 3 2 1



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
372S0186	372S0185		ALL	Alternate Temp Diode

SYNC_MASTER=BRANCH_JERRYCHOW SYNC_DATE=09/10/2014

PAGE TITLE

SENSORS: Temperature Sensors

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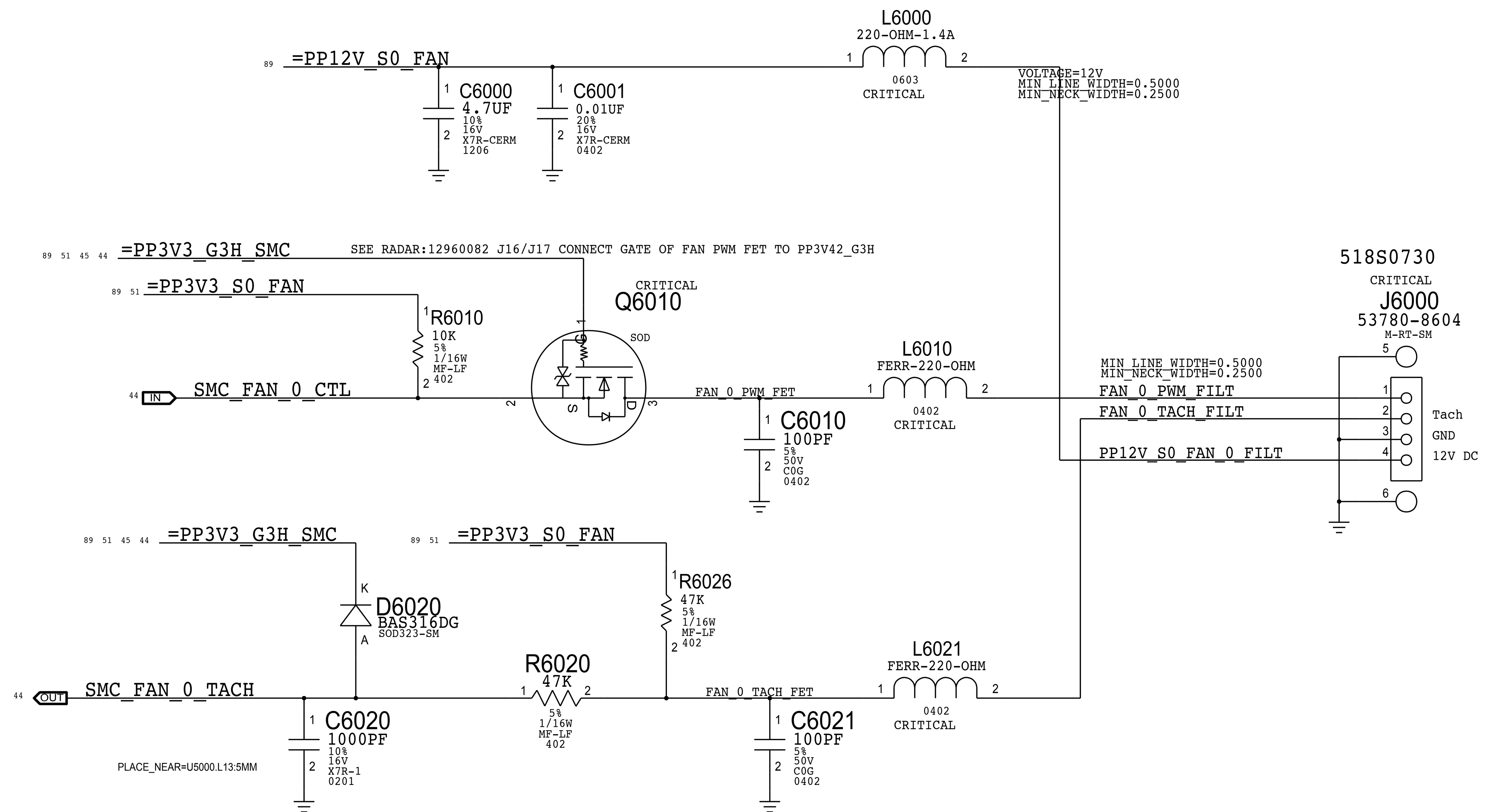
8 7 6 5 4 3 2 1

SMC Fan 0 (System)

Note:
The circuit for the PWM input to the fan acts as a non-inverting level-shifter to protect the SMC. It is assumed there is a pull-up to 5V/12V inside the fan, otherwise when the SMC PWM goes low and Q6010 turns on, there would be 5V/12V present on the SMC pin! Then by definition, the drain of Q6010 is at common and the SMC sinks current when Q6010 is on.

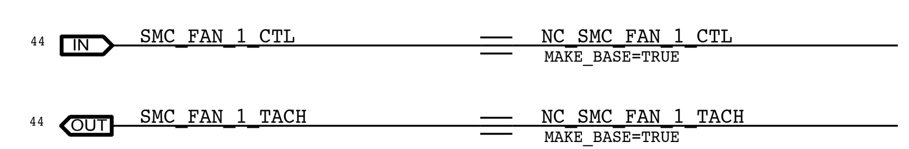
This resembles an open-drain if there is a pull-up, going to a Hi-Z FET input.

Otherwise, this is simply a pass-FET.
See RADAR: 10565825- D7: Need scematic and PCB file of fan(All Vendors).



Add C6020 1000pF Cap, Change R6020 to 47K -- Radar 11661918 D8 Proto1 Fan Tach instability.

SMC Fan 1 (Unused)



SYNC_MASTER=J16_IG SYNC_DATE=04/29/2013

PAGE TITLE
FAN: System Fan

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SYNC_MASTER=J17_DIRK SYNC_DATE=03/07/2013

PAGE TITLE

AUDIO: CODEC/REGULATORS

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SYNC_MASTER=J78_DAVID

SYNC_DATE=11/18/2013

PAGE TITLE

AUDIO: HEADPHONE AMP

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SIZE D

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LEFT CH SPEAKER AMP

APPLE P/N 353S3163

SPEAKER AMP GAIN = +12 DB
SPEAKER AMP RIN = 40K NOMINAL
FC_HPF, TWEETERS = ~847 HZ (4700 PF)
FC_HPF, WOOFERS = ~4 HZ (1.0 UF)

EDGE RATE

CONTROL	R6304	R6305
ON	0 OHM	NOSTUFF
OFF	NOSTUFF	0 OHM

AUD_RAMP_MONO NET:
HIGH = MONO OPERATION
LOW = STEREO OPERATION

GAIN	R6306	R6307
+9 DB	NOSTUFF	0 OHM
+12 DB	NOSTUFF	47 KOHM
+15 DB	NOSTUFF	NOSTUFF
+18 DB	47 KOHM	NOSTUFF
+24 DB	0 OHM	NOSTUFF

SYNC_MASTER=J78_DAVID SYNC_DATE=11/18/2013

PAGE TITLE: AUDIO: LEFT SPKR AMP

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BRANCH

RIGHT CH SPEAKER AMP

APPLE P/N 353S3163

SPEAKER AMP GAIN = +12 DB
SPEAKER AMP RIN = 40K NOMINAL
FC_HPF, TWEETERS = ~847 HZ (4700 PF)
FC_HPF, WOOFERS = ~4 HZ (1.0 UF)

EDGE RATE CONTROL	R6404	R6405
ON	0 OHM	NOSTUFF
OFF	NOSTUFF	0 OHM

AUD_RAMP_MONO NET:
HIGH = MONO OPERATION
LOW = STEREO OPERATION

GAIN	R6406	R6407
+9 DB	NOSTUFF	0 OHM
+12 DB	NOSTUFF	47 KOHM
+15 DB	NOSTUFF	NOSTUFF
+18 DB	47 KOHM	NOSTUFF
+24 DB	0 OHM	NOSTUFF

SYNC_MASTER=J78_DAVID SYNC_DATE=11/18/2013

PAGE TITLE: AUDIO: RIGHT SPKR AMP

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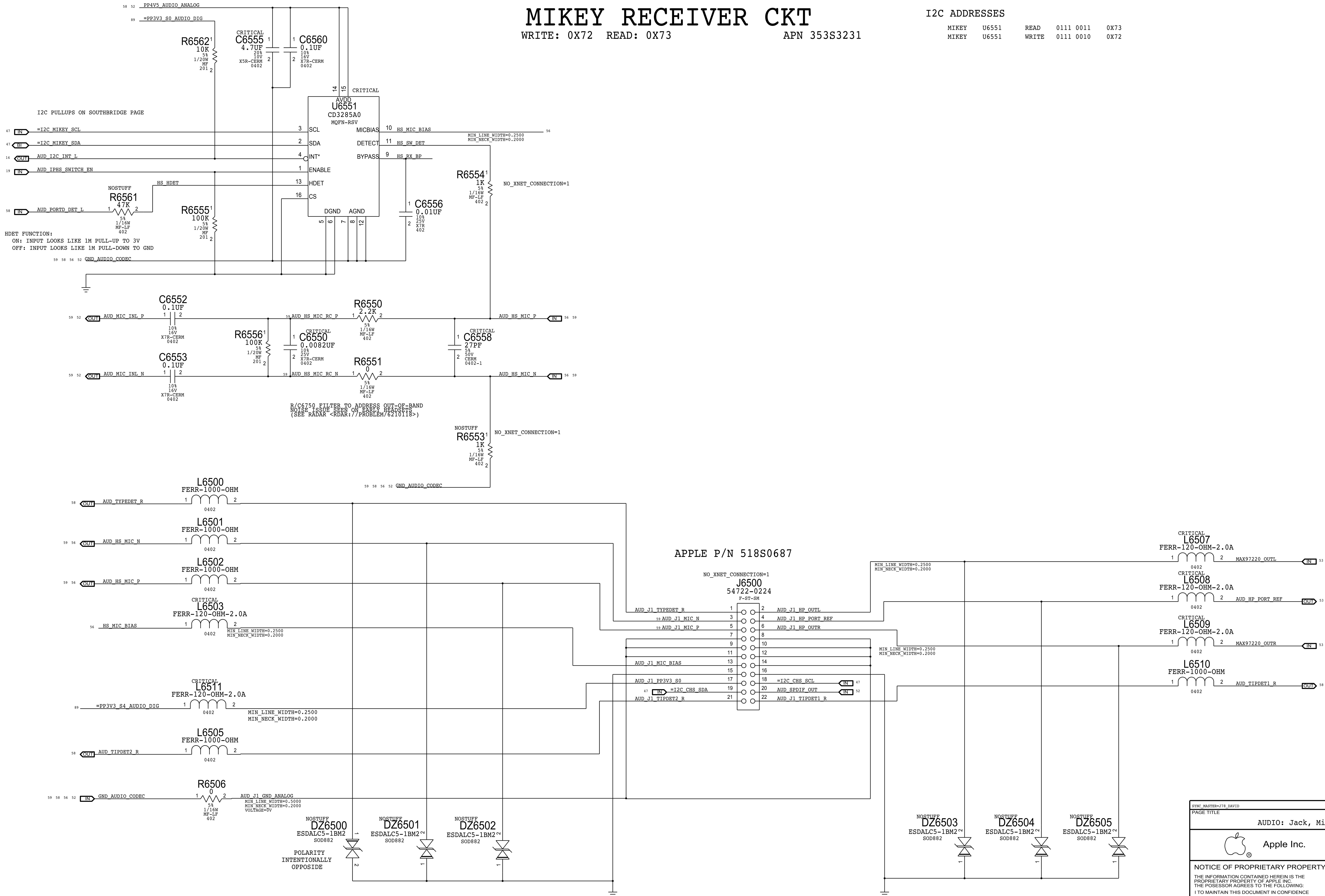
MIKEY RECEIVER CKT

WRITE: 0X72 READ: 0X73

APN 353S3231

I2C ADDRESSES

MIKEY	U6551	READ	0111 0011	0X73
MIKEY	U6551	WRITE	0111 0010	0X72



SYNC_MASTER=J78_DAVID SYNC_DATE=11/18/2013

PAGE TITLE

AUDIO: Jack, Mikey, CHS Switch

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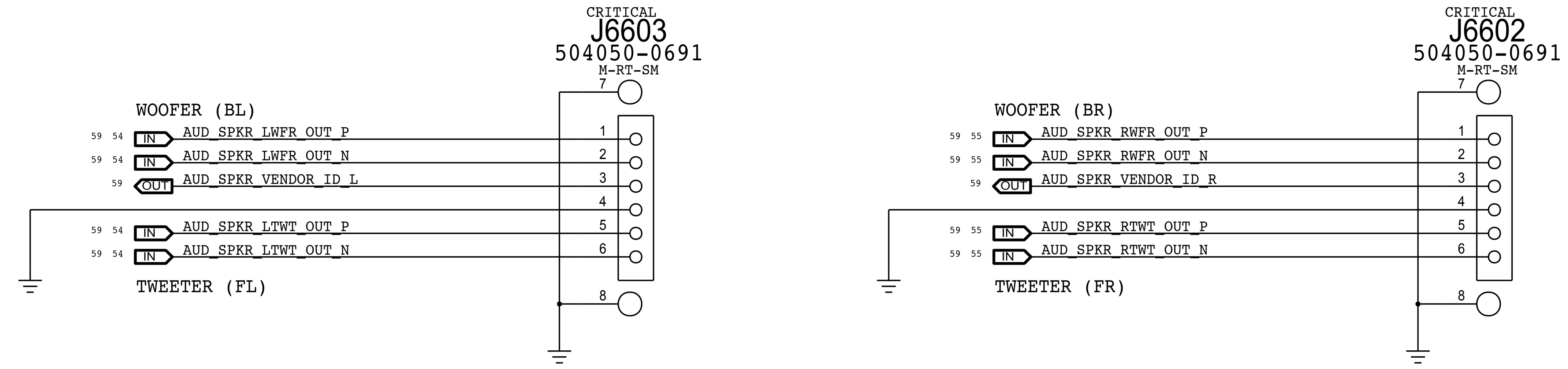
SPEAKER CABLE CONNECTORS

APPLE P/N 518S0862



SYNC_MASTER=J17_DIRK	SYNC_DATE=03/07/2013
PAGE TITLE: AUDIO: Spkr/Mic Conn.	
Apple Inc.	DRAWING NUMBER: 051-00673 SIZE: D
	REVISION: 0.24.0
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IPHS HS Detect Debounce CKT

AUDIO CONNECTOR DETECT STATES

	NOTHING	SPDIF	HEADPHONE
AUD_TYPEDET_R	1	1	0
AUD_TIPDET*_R	0	1	1
AUD_OUTJACK_INSERT	0	1	1
AUD_SENSE_A	1	20K/2.67K RDIV	5.11K/2.67K RDIV

SYNC_MASTER=J17_DIRK SYNC_DATE=03/07/2013

PAGE TITLE

AUDIO: Detects/Grounding

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CODEC OUTPUT SIGNAL PATHS

FUNCTION	VOLUME/MUTE	CONVERTER	PIN COMPLEX	MAC SHDN	WIN SHDN	DET ASSIGNMENT
HP/LINE OUT	0X03 (3)	0X03 (3)	0X0A (10,D)	GPIO_2	GPIO_2	0X0A (DET D)
PRIMARY SPKRS (WFR)	0X04 (4)	0X04 (4)	0X0B (11)	MICBIAS	GPIO_3	N/A
SECONDARY SPKRS (TWT)	0X03 (3)	0X03 (3)	0X0A (10,V24)	MICBIAS	N/A	N/A
SPDIF OUT	N/A	0X08 (8)	0x10 (16)	N/A	N/A	0X0D (DET B)

CODEC INPUT SIGNAL PATHS

FUNCTION	CONVERTER	PIN COMPLEX	ENABLE/CONTROL	DET ASSIGNMENT
SPDIF IN	0X07 (7)	0x0F (15)	N/A	0X09 (DET A)
INTERNAL MIC ARRAY	0X06 (6) 0X05 (5)	0X0E (14,LEFT & RIGHT) 0X12 (18,LEFT)	N/A	N/A
EXTERNAL MIC	0X06 (6)	0X0D (13,V22,B,LEFT)	Lynx POINT GPIO 16	Lynx POINT GPIO 5 (RCVR INT) Lynx POINT GPIO 3 (PERIPH DET)

OTHER DETECT

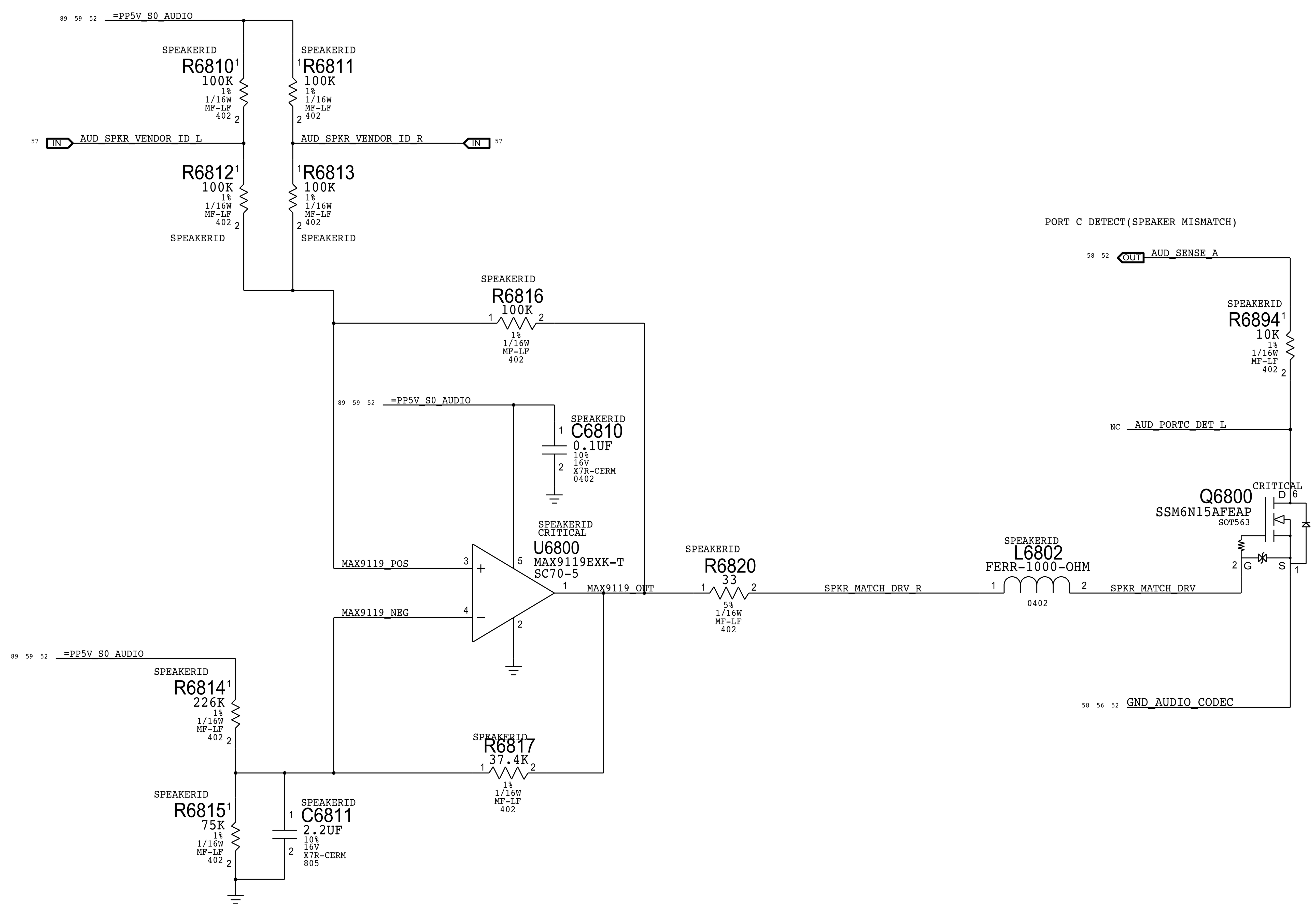
FUNCTION	CONVERTER	PIN COMPLEX	ENABLE/CONTROL	DET ASSIGNMENT
MULTIPLE SPKR VENDORS	N/A	N/A	N/A	0X0C (DET C)

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
AUDIO	*	0.1 MM	?
SPKROUT	*	0.2 MM	?

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
AUDIODIFF	*	Y	0.1 MM	0.1 MM	10 MM	0.1 MM	0.1 MM
SPKROUTDIFF	*	Y	0.6 MM	0.25 MM	10 MM	0.2 MM	0.2 MM

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
AUDIODIFF	*	AUDIODIFF
SPKROUTDIFF	*	SPKROUTDIFF

CIRCUIT THEORY OF OPERATION AVAILABLE IN <RDAR://PROBLEM/9776522>



ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
I215 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD LO1 L P	52 53 54
I216 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD LO1 L N	52 53 54
I217 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD LO1 L C P	53
I218 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD LO1 L C N	53
I211 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD LO1 R P	52 53 55
I212 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD LO1 R N	52 53 55
I210 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD LO1 R C P	53
I209 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD LO1 R C N	53
I208 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD LO2 L P	52 54
I206 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD LO2 L N	52 54
I207 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD LO2 R P	52 55
I204 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD LO2 R N	52 55
I205 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD RAMP LINC P	55
I203 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD RAMP LINC N	55
I220 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD RAMP RINC P	55
I219 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD RAMP RINC N	55
I222 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD RAMP LIN P	55
I221 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD RAMP LIN N	55
I224 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD RAMP RIN P	55
I223 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD RAMP RIN N	55
I226 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD LAMP LINC P	54
I225 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD LAMP LINC N	54
I227 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD LAMP RINC P	54
I229 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD LAMP RINC N	54
I228 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD LAMP LIN P	54
I230 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD LAMP LIN N	54
I264 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD LAMP RIN P	54
I263 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD LAMP RIN N	54
I236 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220 INL P	53
I238 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220 INL N	53
I237 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220 INR P	53
I239 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220 INR N	53
I240 SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD SPKR RWFR OUT P	55 57
I241 SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD SPKR RWFR OUT N	55 57
I242 SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD SPKR RTWT OUT P	55 57
I243 SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD SPKR RTWT OUT N	55 57
I244 SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD SPKR LWFR OUT P	54 57
I246 SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD SPKR LWFR OUT N	54 57
I245 SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD SPKR LTWT OUT P	54 57
I247 SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD SPKR LTWT OUT N	54 57
I326 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD MIC INL P	52 56
I327 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD MIC INL N	52 56
I328 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD HS MIC RC P	56
I329 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD HS MIC RC N	56
I254 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD HS MIC P	56
I255 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD HS MIC N	56
I324 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD J1 MIC P	56
I325 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD J1 MIC N	56

SYNC_MASTER=J78_DAVID SYNC_DATE=11/21/2013

PAGE TITLE
AUDIO: Speaker ID

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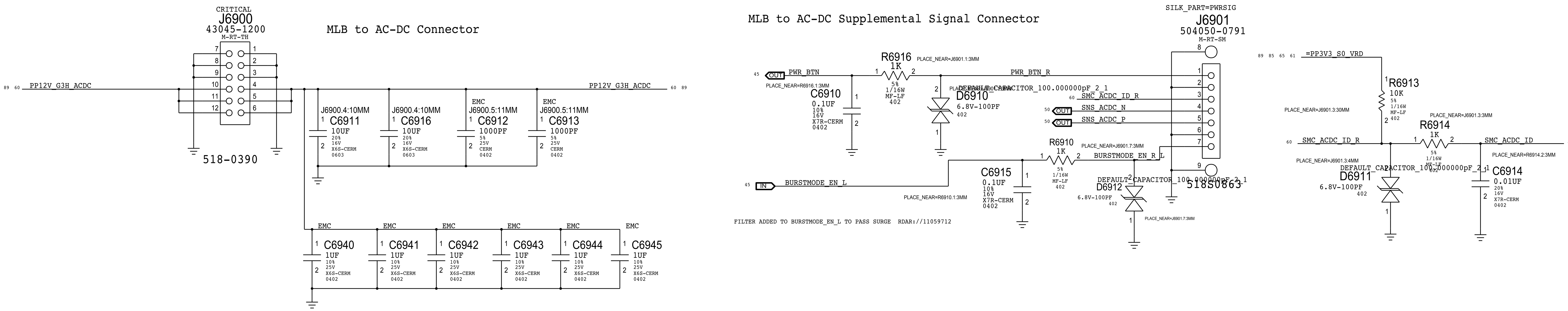
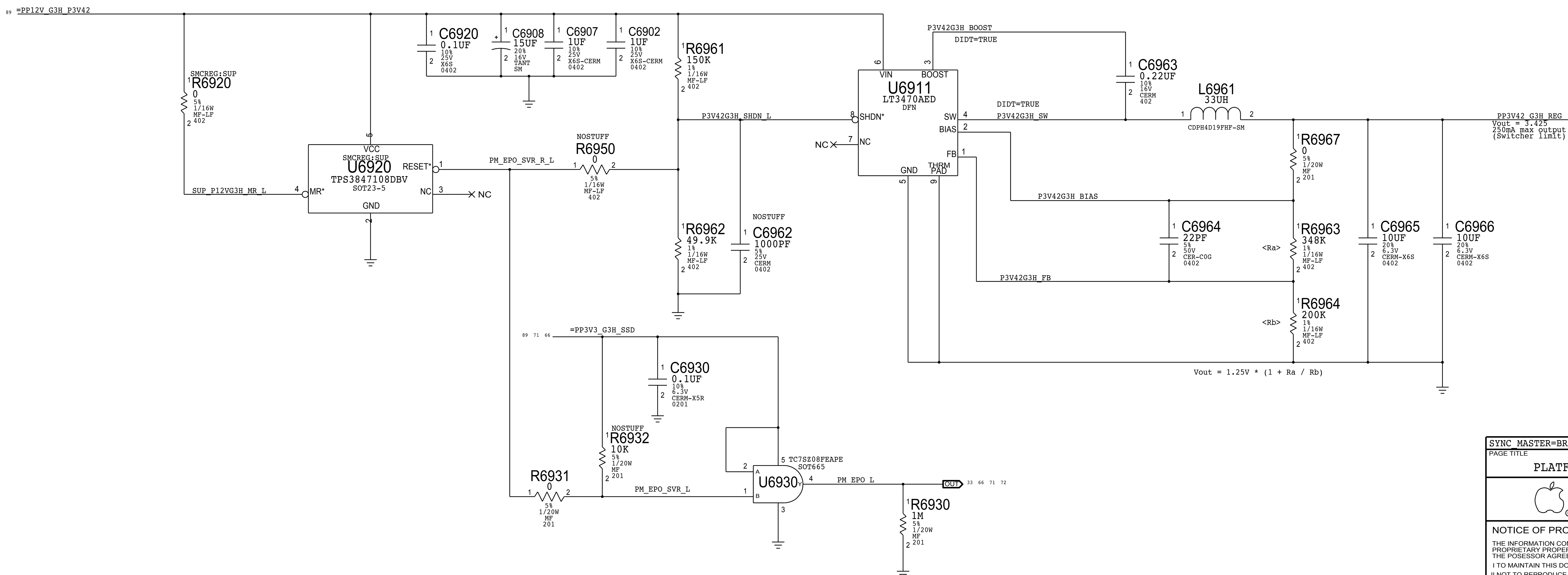
MLB to AC-DC Connector

MLB to AC-DC Supplemental Signal Connector



3.425V "G3Hot" Regulator

Max avg current: 0.04 A (BUDGET)
Max peak current: 0.10 A (BUDGET)



SYNC_MASTER=BRANCH_HCHENG SYNC_DATE=10/29/2014

PAGE TITLE

PLATFORM POWER: Connectors / VReg G3Hot

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051-00673	D

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SYNC_MASTER=BRANCH_HCHENG SYNC_DATE=10/29/2014

PAGE TITLE

CPU & CHIPSET: CPU CORE VR

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DRAWING NUMBER 051-00673 SIZE D

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DRAWING	LAST_MODIFIED=Thu Feb 12 14:48:21 2015
SYNC_MASTER=J95_HARPER	SYNC_DATE=02/11/2015
PAGE TITLE	CPU & CHIPSET: CPU CORE VR (VCC)
DRAWING NUMBER	051-00673
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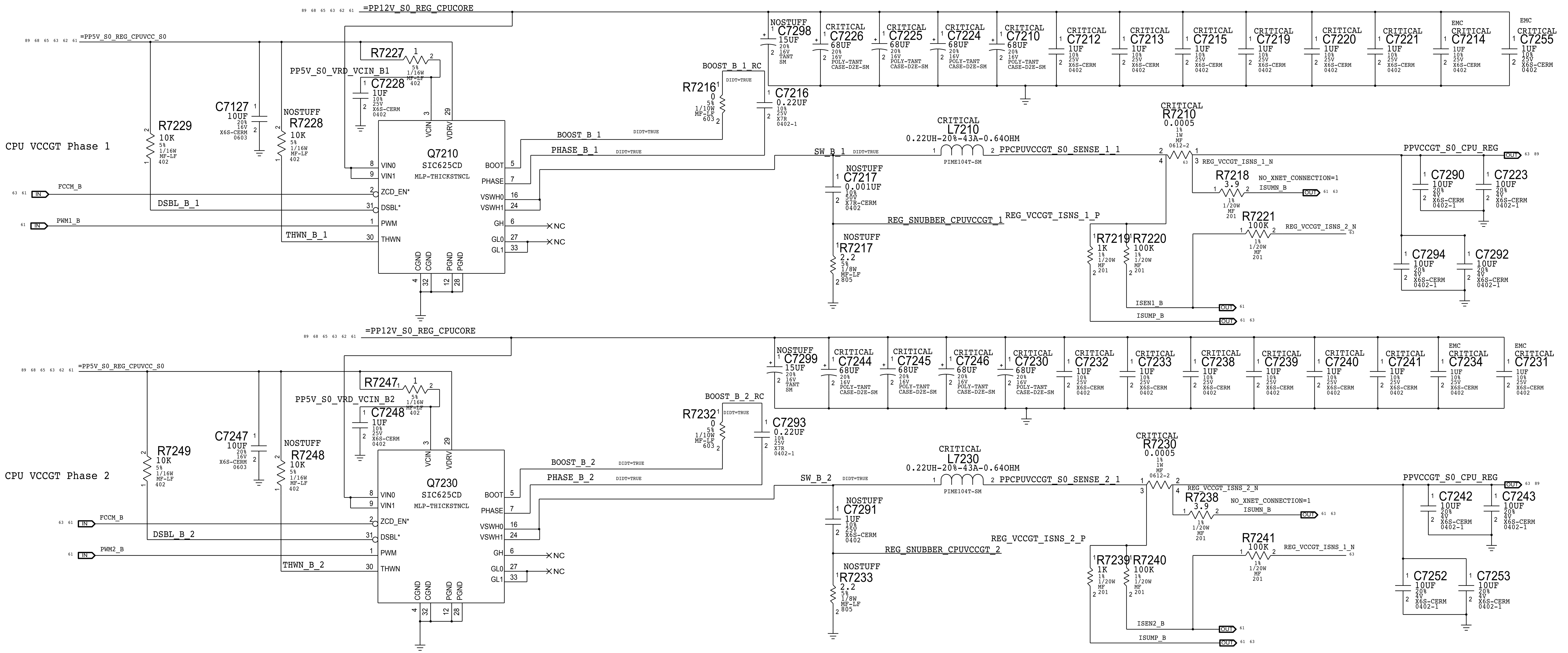
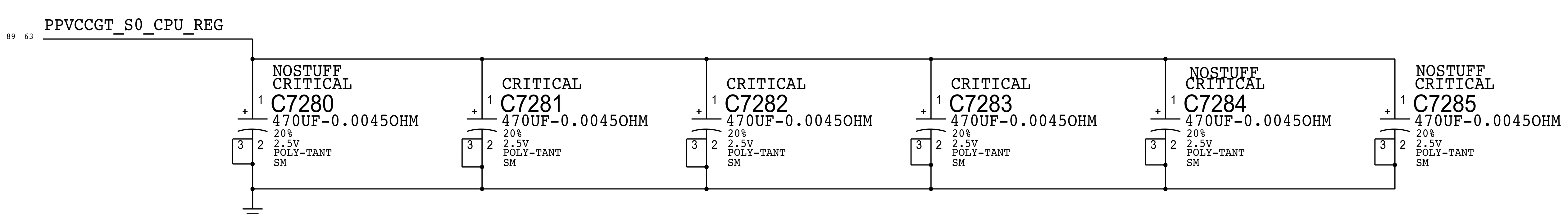
CPU VCCGT Regulator

EDC = 51A

TDC = 37A



CPU VCCGT OUTPUT DECOUPLING



DRAWING	LAST_MODIFIED=Thu Feb 12 14:48:21 2015
SYNC_MASTER=J95_HARPER	SYNC_DATE=02/11/2015
PAGE TITLE	CPU & CHIPSET: CPU CORE VR (VCCGT)
Apple Inc.	DRAWING NUMBER 051-00673, SIZE D
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NOTICE OF PROPRIETARY PROPERTY:

VDDQ (1.35V) S3 REGULATOR

EDC = 20A

TDC = 14A

SYNC_MASTER=BRANCH_HCHENG SYNC_DATE=10/29/2014

PAGE TITLE

CPU & CHIPSET: CPU VDDQ VR

Apple Inc.

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VCCIO (0.95V) S0 REGULATOR

EDC = 5.5A

TDC = 5.5A

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
12880358	2	CAP,470uF,0.006OHM,2V,D2	C7540,C7541	VR_BULKCAP:CURRENT
12880381	2	CAP,470uF,0.0045OHM,2.5V,SM	C7540,C7541	VR_BULKCAP:FUTURE

DRAWING LAST_MODIFIED=Thu Feb 12 14:48:22 2015

SYNC_MASTER=J78_MLB SYNC_DATE=08/02/2014

PAGE TITLE

CPU & CHIPSET: CPU VCCIO VR

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SYNC_MASTER=J95_HARPER SYNC_DATE=02/11/2015

PAGE TITLE: PLATFORM POWER: 3.3V S5/5V S4 VR

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DRAWING NUMBER: 051-00673 SIZE: D

REVISION: 0.24.0

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PAGE: 76 OF 121

SHEET: 66 OF 93

SYNC_MASTER=J95_HARPER SYNC_DATE=02/11/2015

PAGE TITLE

CPU & CHIPSET: PCH 1V0 VR

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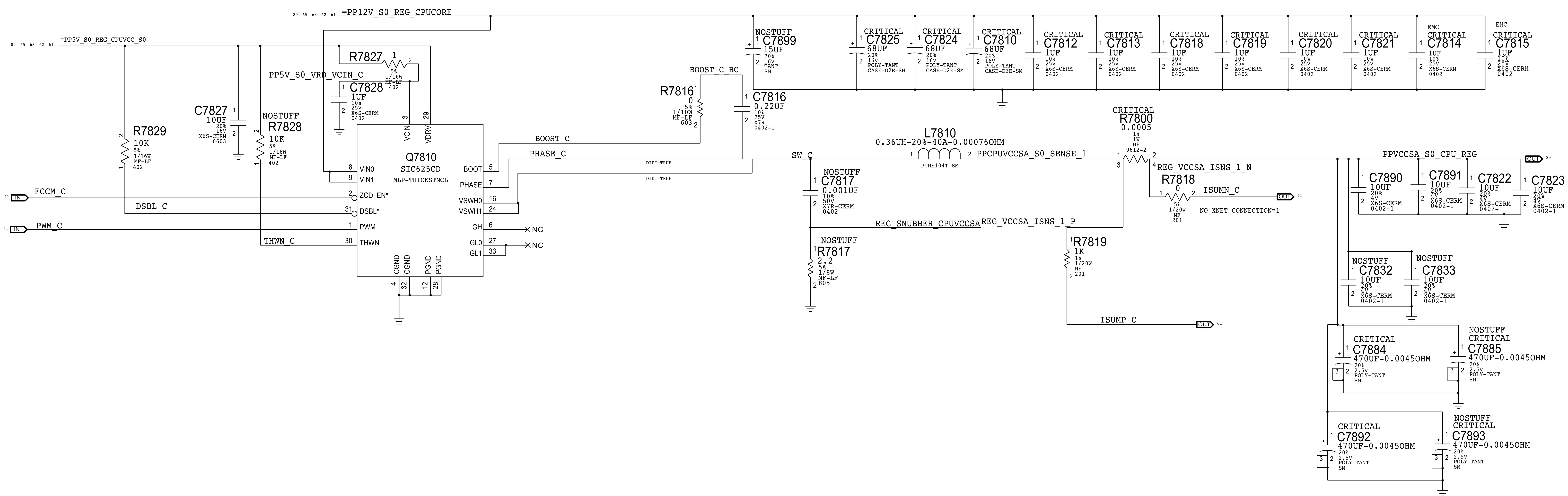
CPU VCCSA Regulator

EDC = 11.1A

TDC = 10A



DRAWING LAST_MODIFIED=Thu Feb 12 14:48:22 2015

SYNC_MASTER=J95_HARPER SYNC_DATE=02/11/2015

PAGE TITLE

CPU & CHIPSET: CPU CORE VR (VCCSA)

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OMIT/table L8100 due to late sourcing change, did not want footprint change
Add 152s1668 (Cyntec) directly for next program

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S1078	138S0810		C8106, C8108	BLC OUTPUT CAPS
371S0748	371S0731		D8103	INPUT DIODE
371S0782	371S0783		D8101	OUTPUT DIODE
376S0845	376S1067		Q8119	POWERLINE FET
376S1204	376S1106		Q8211 TO Q8219	LED SINK FET
740S0145	740S0146		F8100	FUSE FOR BLC
107S0359	107S0229		R8102	BOOST SENSE RESISTOR

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
152S1668	1	IND,PWR,33UH,20%,10A,35MOHM	L8100	CRITICAL	
353S00079	1	IC,PLP8565A13,LED BCKLGHT CTRL,7X7 QFN56	U8100	CRITICAL	
126-0190	2	BOOST POLYMER OUTPUT CAP	C8109, C8118	CRITICAL	

SYNC_MASTER=MLB_CTO SYNC_DATE=05/14/2014

PAGE TITLE

DISPLAY: LCD Backlight Driver (LP8565)

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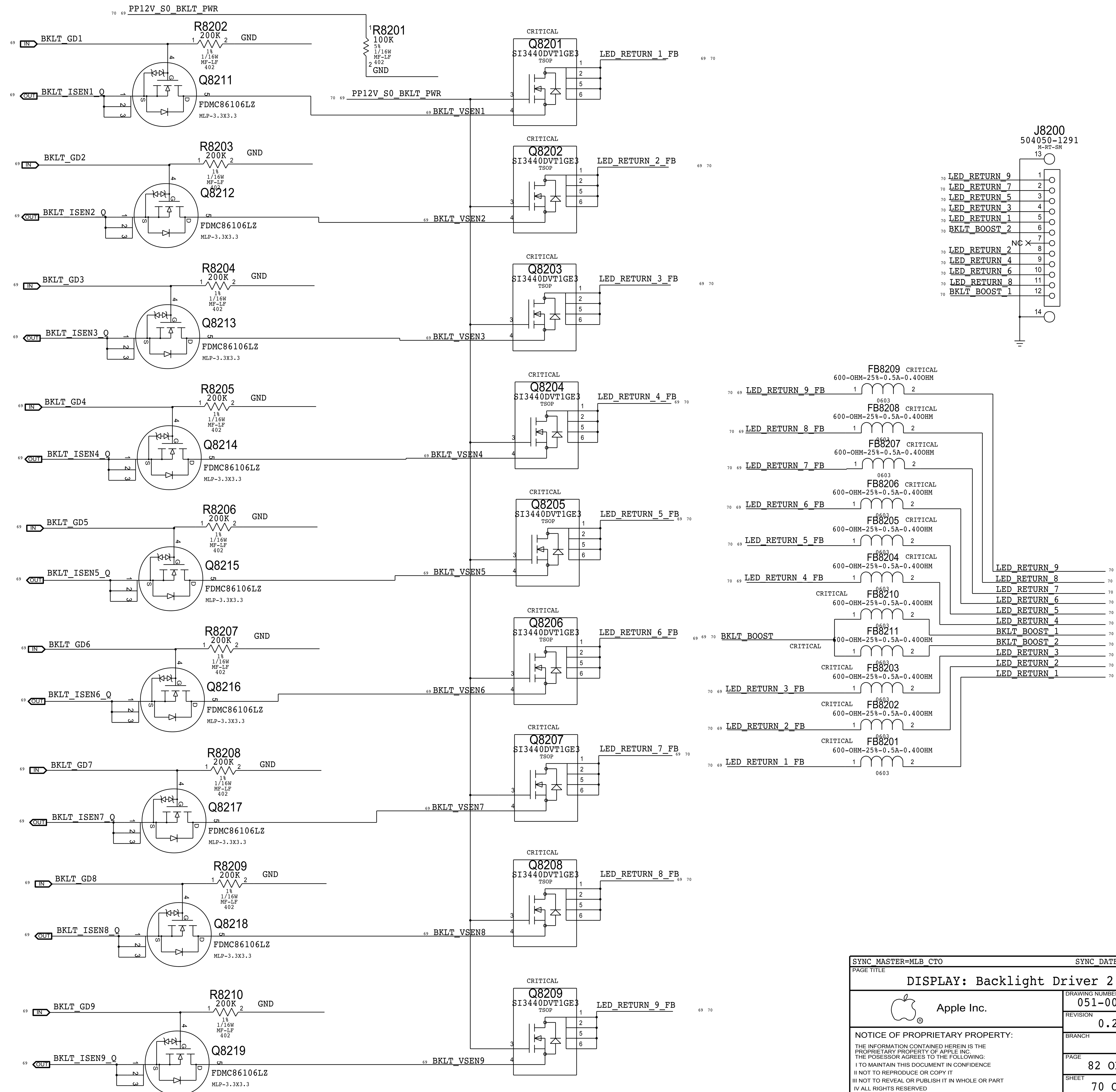
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S1256	376S1073		ALL	Short Protection FET
155S0831	155S0797		ALL	FB8201 TO FB8211



SYNC_MASTER=MLB_CTO	SYNC_DATE=05/14/2014
PAGE TITLE	DISPLAY: Backlight Driver 2
DRAWING NUMBER	051-00673
SIZE	D
REVISION	0.24.0
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SHEET	70 OF 93

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SYNC_MASTER=J17_MAX SYNC_DATE=02/11/2013

PAGE TITLE: PLATFORM POWER: FET-Controlled S0 and S4

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DRAWING NUMBER: 051-00673 SIZE: D

REVISION: 0.24.0

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Power Sequencing requirements

System:

1. 12V_ACDC -> 3V42G3H -> 12V_S5/12V_S5_SSD -> 3V3_S5 -> 1V0_S5 -> 5V_S4/3V3_S4 -> 1V0_S3

Intel CPU:

2. 1V0_S3 -> VDDQ_S3 -> 12V_S0 -> 5V_S0 -> 3V3_S0 -> GPU Rails -> VCCIO_S0-> VCC_CPU_S0 (VCCGT & VCCSA)

AMD GPU:

3. 3V3_S0 -> 0V95 -> 1V8 (VDD_CT) -> GPUCORE (VDDC) -> VDDCI -> FBVDDQ

SYNC_MASTER=J78_KENNY SYNC_DATE=12/18/2013

PAGE TITLE

PLATFORM POWER: Regulator Enables

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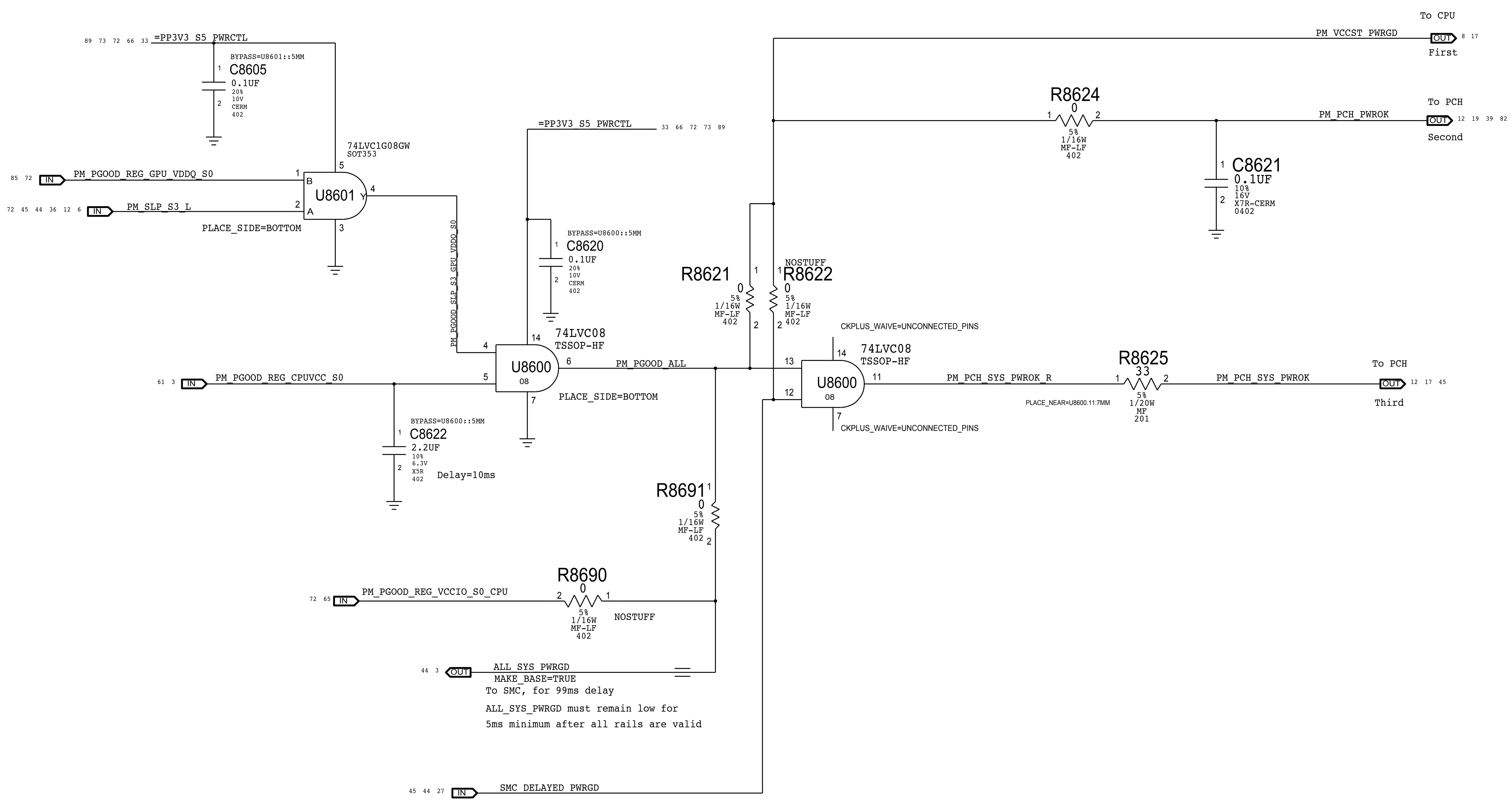
ALL_SYS_PWRGD,PCH_PWROK & SYS_PWROK Generation

PCH Power Goods



Resume Reset

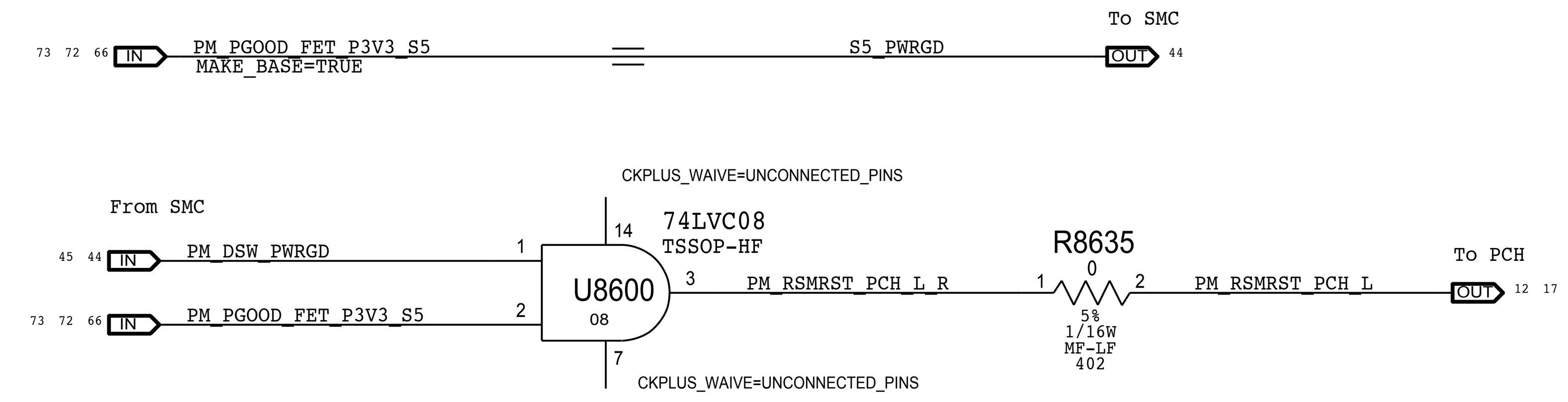
Intel Doc# 29517 Maho Bay PDG, Section 22.13
Intel Doc# 29562 Panther Point EDS, Section 8.7 and 8.8

Note:
THE IMAC J78 DESIGNS DOES NOT SUPPORT DEEP SX MODES SO BOTH DPWROK AND RSMRST# signals are shorted together

Requirements:
- Power on:
  - Asserted at least 10 ms after all suspend well power is valid
- Power off or loss of AC:
  - Transition to 0.8V or less before VccSUS3_3 drops to 2.90 V to allow PCH to switch suspend well to battery without excessive loading

Method:
The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation via PM_DSW_PWRGD.

RSMRST# is asserted when power good from regulator is de-asserted in the event AC is lost. Power good de-assertion should happen quickly enough to meet Intel spec.



SYNC_MASTER=J78_KENNY SYNC_DATE=10/09/2013

PAGE TITLE: PLATFORM POWER: PM Power Good

DRAWING NUMBER: 051-00673 SIZE: D

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Page Notes

POLARITY SWAPS INTENDED ON LANES,SEE NOTES AT DIFF PAIRS.
ALL LANES ARE ALSO REVERSED, SEE ALIASES ON CSA 112

SYNC_MASTER=J95 SYNC_DATE=08/26/2013

PAGE TITLE

GFX: Emerald PCIe

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SYNC_MASTER=J94

SYNC_DATE=08/26/2013

PAGE TITLE

GFX: Emerald CORE/FB POWER

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8 7 6 5 4 3 2 1

D

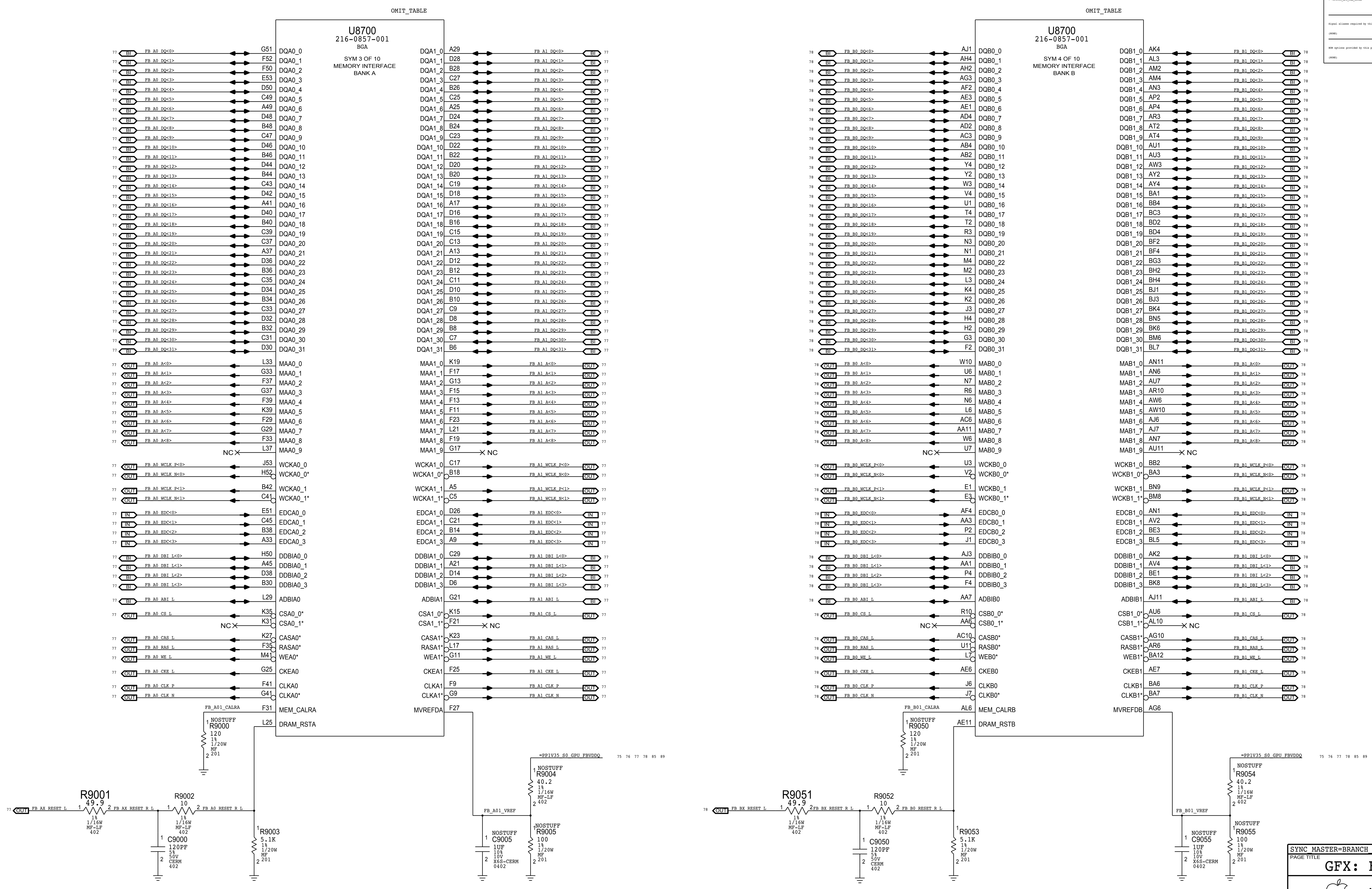
C

B

A

NOTE:GDDR5 MODE H MAPPING

Page Notes



SYNC_MASTER=BRANCH_FIYIN SYNC_DATE=09/18/2014

PAGE TITLE

GFX: Emerald FRAME BUFFER

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8 7 6 5 4 3 2 1

SYNC_MASTER=BRANCH_FIYIN SYNC_DATE=09/18/2014

PAGE TITLE

GDDR5 Frame Buffer A

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Page Notes
Power aliases required by this page:
- =PP1V35_S0_GPU_FBVDDQ
Signal aliases required by this page:
(NONE)
BOM options provided by this page:
(NONE)
NORMAL
U9300
32MX32-1.5GHZ-MFL
BGA
K4G10325FG-HC03
(1 OF 2)
(2 OF 2)
OMIT_TABLE
MIRRORED
U9350
32MX32-1.5GHZ-MFH
=PP1V35_S0_GPU_FBVDDQ
FB_B0_VREFC
FB_B0_VREFD0
FB_B0_VREFD1
FB_B1_VREFC
FB_B1_VREFD0
FB_B1_VREFD1
MIN_LINE_WIDTH=0.2500
MIN_NECK_WIDTH=0.1000
NOSTUFF
VDD
VDDQ
VSS
VSSQ
VREFC
VREFD
SYNC_MASTER=BRANCH_FIYIN
SYNC_DATE=09/18/2014
PAGE TITLE
GDDR5 Frame Buffer B
Apple Inc.
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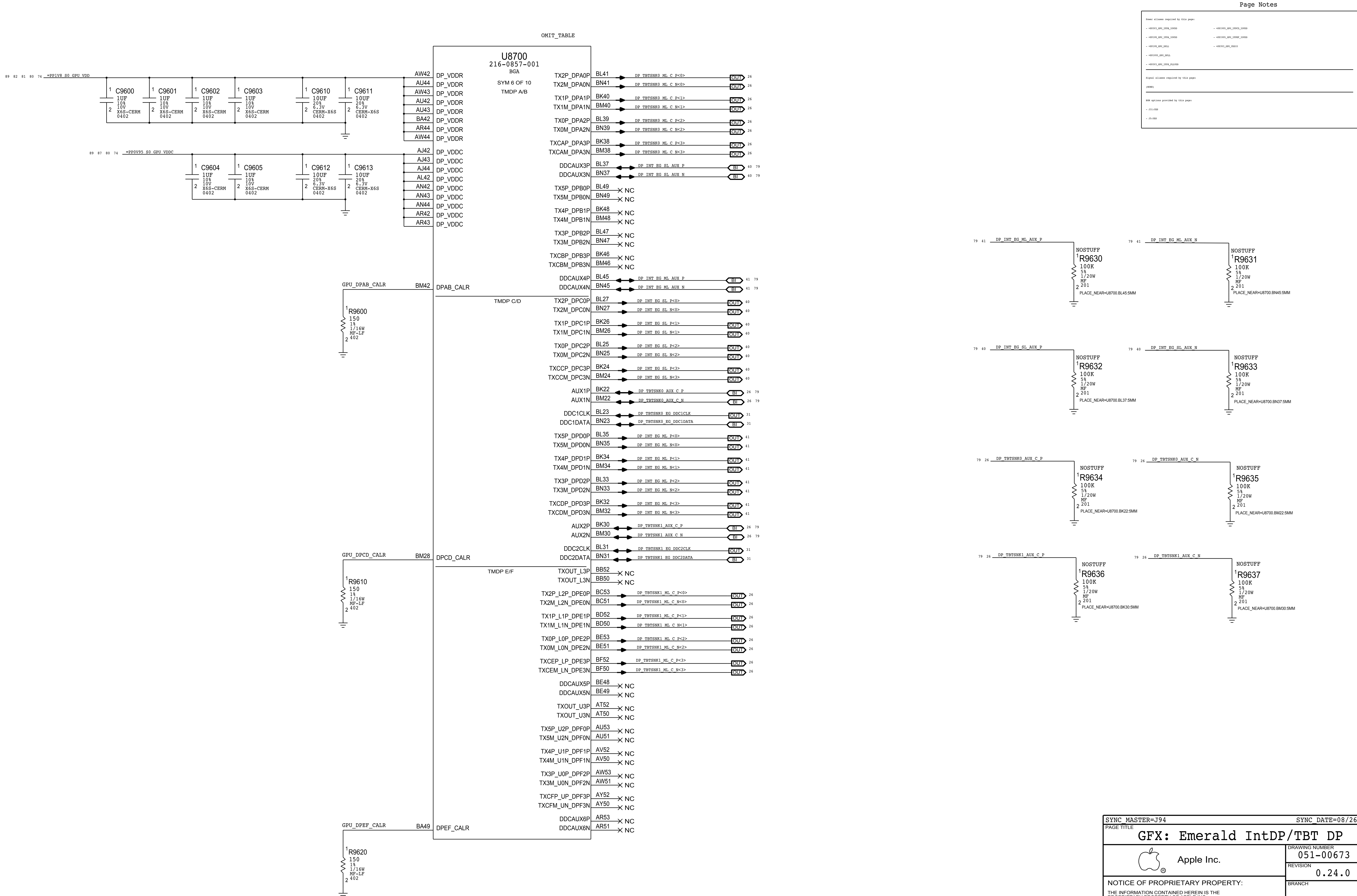

SYNC_MASTER=J94 SYNC_DATE=08/26/2013

PAGE TITLE

GFX: Emerald IntDP/TBT DP

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SYNC_MASTER=J94 SYNC_DATE=08/26/2013

PAGE TITLE

GFX: Emerald GPIOS,CLK,STRAPS

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SYNC_MASTER=J94 SYNC_DATE=08/26/2013

PAGE TITLE

GFX: Emerald Gnds & Unused

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O/P= PPGPUCORE_S0_REG

GPUCORE
VOUT = VCORE
PEAK = 195A
AVG = 154A

SVC	SVD	GPUCORE
0	0	1.1 V
0	1	1.0 V
1	0	0.9 V
1	1	0.8 V

DEFAULT->

DRAWING LAST_MODIFIED=Thu Feb 12 14:48:25 2015

SYNC_MASTER=J78_NAT SYNC_DATE=02/06/2014

PAGE TITLE

GRAPHICS: GPU CORE VR

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SYNC_MASTER=J78_NAT SYNC_DATE=02/13/2014

PAGE TITLE

GRAPHICS: GPU CORE VR (PHASES 1-3)

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PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
128S0198	5	CAP,330UF,0.009OHM,2V,CASE-D2	CA380,CA381,CA382,CA383,CA384	VR_BULKCAP:CURRENT
128S0198	5	CAP,330UF,0.009OHM,2V,CASE-D2	CA385,CA386,CA387,CA388,CA389	VR_BULKCAP:CURRENT
128S0381	5	CAP,470UF,0.0045OHM,2.5V,TANT	CA380,CA381,CA382,CA383,CA384	VR_BULKCAP:FUTURE
128S0381	5	CAP,470UF,0.0045OHM,2.5V,TANT	CA385,CA386,CA387,CA388,CA389	VR_BULKCAP:FUTURE

DRAWING LAST_MODIFIED=Thu Feb 12 14:48:25 2015

SYNC_MASTER=J78_NAT SYNC_DATE=02/13/2014

PAGE TITLE

GRAPHICS: GPU CORE VR (PHASES 4-6)

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GPU VDDQ SUPPLY

VOUT = 1.5V / 1.35V

F = 500 KHZ

EDC = 12A

TDC = 9.5A

Note:
Regulator requires a minimum load to prevent noise in the audio frequencies

	VID 1	VID 0	GPU VDDQ
	1	1	1.35 V
DEFAULT ->	0	1	1.5 V
	0	0	1.6 V
	1	0	1.373 V

XTA: 1.35/1.5
LEA: 1.35
PIT: 1.35

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
114S0269	2	RES,3.4K,1%,1/16W,0402,SMD	RA470, RA471	FB:4G_HYNIX
114S0271	2	RES,3.57K,1%,1/16W,0402,SMD	RA470, RA471	FB:4G_ELPIDA

DRAWING LAST_MODIFIED=Thu Feb 12 14:48:25 2015

SYNC_MASTER=J78_MLB SYNC_DATE=02/20/2014

PAGE TITLE

GRAPHICS: GPU VDDQ VR

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NEPTUNE

	VID 1	VID 0	GPU VDDCI
DEFAULT ->	1	1	0.850 V
	1	0	1.000 V

AMETHYST

	VID 1	VID 0	GPU VDDCI
DEFAULT ->	1	1	0.900 V
	1	0	1.000 V

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
114S0363	1	RES,31.6K,0402	RA574	GPU_Emerald
116S0068	2	RES,1.2K,0402	RA570,RA571	GPU_Emerald
114S0378	1	RES,45.3K,0402	RA575	GPU_Emerald
114S0363	1	RES,31.6K,0402	RA574	GPU_Neptune
114S0231	2	RES,1.4K,0402	RA570,RA571	GPU_Neptune
114S0347	1	RES,22.1K,0402	RA575	GPU_Neptune
114S0373	1	RES,40.2K,0402	RA574	GPU_AmethystP_XTA
114S0228	2	RES,1.3K,0402	RA570,RA571	GPU_AmethystP_XTA
114S0347	1	RES,22.1K,0402	RA575	GPU_AmethystP_XTA
114S0382	1	RES,48.7K,0402	RA574	GPU_AmethystP_PROA
114S0228	2	RES,1.3K,0402	RA570,RA571	GPU_AmethystP_PROA
114S0347	1	RES,22.1K,0402	RA575	GPU_AmethystP_PROA

SYNC_MASTER=hcheng_j95 SYNC_DATE=02/10/2015

PAGE TITLE

GRAPHICS: GPU VDDCI VR

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GPU VDDC (0.95V) S0 REGULATOR

Max avg current: 4.3 A
Max peak current: 4.5 A
Switching freq: 300 kHz

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
128S0358	2	CAP,470uF,0.006OHM,2V,D2	CA640,CA641	VR_BULKCAP:CURRENT
128S0381	2	CAP,470uF,0.0045OHM,2.5V,SM	CA640,CA641	VR_BULKCAP:FUTURE

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SYNC_MASTER=J78_MLB SYNC_DATE=08/02/2014

PAGE TITLE
GRAPHICS: GPU 0V95 VR

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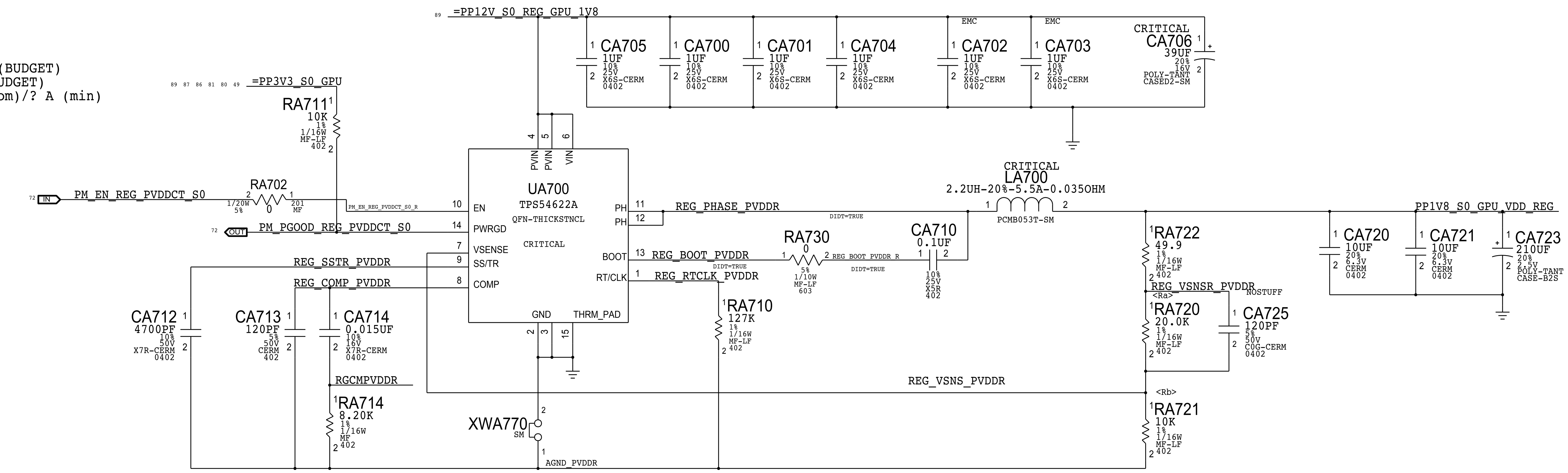
1.8V S0 Regulator

Max avg current: 2.4 A (BUDGET)
Max peak current: 3 A (BUDGET)
OC trip point: ? A (nom)/? A (min)
Switching freq: ? kHz



VOUT = 0.6 * (1 + RA / RB)

DRAWING LAST_MODIFIED=Thu Feb 12 14:48:26 2015
SYNC_MASTER=J95_HARPER SYNC_DATE=02/11/2015

PAGE TITLE: GRAPHICS: GPU 1V8 VR

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SYNC_MASTER=J78_MLB SYNC_DATE=11/20/2013

PAGE TITLE

Power Connectors/Aliases

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PEG aliases

74 PEG_D2R_N<0..15> MAKE_BASE=TRUE = =PEG_D2R_N<15..0> 5
74 PEG_D2R_P<0..15> MAKE_BASE=TRUE = =PEG_D2R_P<15..0> 5
74 PEG_R2D_C_N<0..15> MAKE_BASE=TRUE = =PEG_R2D_C_N<15..0> 5
74 PEG_R2D_C_P<0..15> MAKE_BASE=TRUE = =PEG_R2D_C_P<15..0> 5

GPU ALIASES

80 74 GPU_RESET_L MAKE_BASE=TRUE = TP_GPU_RESET_L 19
82 VR_GPU_RESET_L MAKE_BASE=TRUE = TP_VR_GPU_RESET_L 19

GPU VDDCI PGOOD

72 TP_PM_EN_REG_GPU_VDDCI_S0 = PM_EN_REG_GPU_VDDCI_S0 MAKE_BASE=TRUE 86

SYNC_MASTER=J78_MLB SYNC_DATE=12/05/2013

PAGE TITLE
Signal Aliases

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CPU Reserved

17 6 CPU_CFG<15..12> = TP_CPU_CFG<15..12> MAKE_BASE=TRUE

40 DP_INT_PIN_57 = NC_DP_INT_PIN_57 MAKE_BASE=TRUE
40 DP_INT_PIN_55 = NC_DP_INT_PIN_55 MAKE_BASE=TRUE

12 TP_CLINK_DATA = CLINK_DATA MAKE_BASE=TRUE NO_TEST=1
12 TP_CLINK_CLK = CLINK_CLK MAKE_BASE=TRUE NO_TEST=1
12 TP_CLINK_RESET_L = CLINK_RESAT_L MAKE_BASE=TRUE NO_TEST=1
TP_GPU_CB_REFCKN_OUT1 = GPU_CB_REFCKN_OUT1 MAKE_BASE=TRUE
TP_GPU_CB_REFCKP_OUT1 = GPU_CB_REFCKP_OUT1 MAKE_BASE=TRUE
TP_GPU_PLL_ANALOG_IN = GPU_PLL_ANALOG_IN MAKE_BASE=TRUE
12 TP_PCH_SLP_LAN_L = PCH_SLP_LAN_L MAKE_BASE=TRUE NO_TEST=1
11 TP_PCH_2 = PCH_2 MAKE_BASE=TRUE NO_TEST=1
12 TP_PM_SLP_A_L = PM_SLP_A_L MAKE_BASE=TRUE NO_TEST=1
26 TP_TBT_MONDC1 = TBT_MONDC1 MAKE_BASE=TRUE
26 TP_TBT_PCIE_RESET0_L = TBT_PCIE_RESET0_L MAKE_BASE=TRUE NO_TEST=1

PCH Clocks

TP_ITPXDP_CLK100MP = ITPXDP_CLK100M_P MAKE_BASE=TRUE NO_TEST=1 11 17
TP_ITPXDP_CLK100MN = ITPXDP_CLK100M_N MAKE_BASE=TRUE NO_TEST=1 11 17

PCH Miscellaneous

11 TP_HDA_SDIN1 = NC_HDA_SDIN1 MAKE_BASE=TRUE NO_TEST=1

11 PEG_CLKREQ_L = NC_PEG_CLKREQ_L MAKE_BASE=TRUE
80 14 GPU_NEPTUNE_EMERALD_ID = NC_GPU_IS_AMETHYST MAKE_BASE=TRUE NO_TEST=1

SYNC_MASTER=J78_MLB SYNC_DATE=12/05/2013

PAGE TITLE

Unused Signal Aliases

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8 7 6 5 4 3 2 1

J17 BOARD SPECIFIC PHYSICAL AND SPACING CONSTRAINTS

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP,ISL2,ISL3,ISL4,ISL5,ISL6,ISL7,ISL8,ISL9,ISL10,ISL11,BOTTOM	NO_TYPE,BGA,BGA_TBT,BGA_VRAM	MM	16.2

General Physical Rule Definitions

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	0.1 MM	0.070 MM	10 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	10 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
34_OHM_SE	*	Y	0.185 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
34_OHM_SE	ISL5,ISL8	Y	0.205 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
34_OHM_SE	TOP,BOTTOM	Y	0.220 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
39_OHM_SE	*	Y	0.150 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
39_OHM_SE	ISL5,ISL8	Y	0.165 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
39_OHM_SE	TOP,BOTTOM	Y	0.175 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
42_OHM_SE	*	Y	0.130 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
42_OHM_SE	ISL5,ISL8	Y	0.145 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
42_OHM_SE	TOP,BOTTOM	Y	0.155 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE	*	Y	0.115 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
45_OHM_SE	ISL5,ISL8	Y	0.126 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
45_OHM_SE	TOP,BOTTOM	Y	0.135 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	*	Y	0.090 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
50_OHM_SE	ISL5,ISL8	Y	0.100 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
50_OHM_SE	TOP,BOTTOM	Y	0.105 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	*	Y	0.075 MM	0.075 MM	=STANDARD	=STANDARD	=STANDARD
55_OHM_SE	ISL5,ISL8	Y	0.080 MM	0.080 MM	=STANDARD	=STANDARD	=STANDARD
55_OHM_SE	TOP,BOTTOM	Y	0.085 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
68_OHM_DIFF	*	Y	0.171 MM	0.085 MM	=STANDARD	0.130 MM	0.1 MM
68_OHM_DIFF	TOP,BOTTOM	Y	0.185 MM	0.085 MM	=STANDARD	0.150 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	*	Y	0.136 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM
80_OHM_DIFF	TOP,BOTTOM	Y	0.141 MM	0.085 MM	=STANDARD	0.185 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	Y	0.121 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM
85_OHM_DIFF	TOP,BOTTOM	Y	0.125 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	Y	0.109 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM
90_OHM_DIFF	TOP,BOTTOM	Y	0.111 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	Y	0.086 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM
100_OHM_DIFF	TOP,BOTTOM	Y	0.090 MM	0.085 MM	=STANDARD	0.230 MM	0.1 MM

BGA

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BGA_P1MM	*	=STANDARD	?

Power and Common

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_ISO	*	=STANDARD	8000
GND_P2MM	*	=2:1_SPACING	1000
PWR_P2MM	*	=2:1_SPACING	1100

GENERIC

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GENERIC_ISO	*	=1:1_SPACING	?

BGA Area Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	BGA_P1MM

General Spacing Definitions

Default

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?

Fixed and Dielectric

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1:1_SPACING	*	0.1 MM	?
1X_DIELECTRIC	TOP,BOTTOM	0.071 MM	?
1X_DIELECTRIC	ISL3,ISL10	0.101 MM	?
1X_DIELECTRIC	*	0.076 MM	?

Board Stack-up

FINISHED BOARD THICKNESS: 1.94 MM

Top	Signal	0.5 oz (Cu plated)
	Prepreg	0.071 MM
2	Plane	1 oz
	Core	0.101 MM
3	Signal	0.5 oz
	Prepreg	0.115 MM
4	Plane	1 oz
	Core	0.076 MM
5	Signal	0.5 oz
	Prepreg	0.380 MM
6	Plane	1 oz
	Core	0.076 MM
7	Plane	1 oz
	Prepreg	0.380 MM
8	Signal	0.5 oz
	Core	0.076 MM
9	Plane	1 oz
	Prepreg	0.115 MM
10	Signal	0.5 oz
	Core	0.101 MM
11	Plane	1 oz
	Prepreg	0.071 MM
Btm	Signal	0.5 oz (Cu plated)

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J95 RULE DEFINITIONS

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J17 BOARD SPECIFIC PHYSICAL AND SPACING CONSTRAINTS

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP,ISL2,ISL3,ISL4,ISL5,ISL6,ISL7,ISL8,ISL9,ISL10,ISL11,BOTTOM	NO_TYPE,BGA,BGA_TBT,BGA_VRAM	MM	16.2

General Physical Rule Definitions

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	0.1 MM	0.070 MM	10 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	10 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
34_OHM_SE	*	Y	0.185 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
34_OHM_SE	ISL5,ISL8	Y	0.205 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
34_OHM_SE	TOP,BOTTOM	Y	0.220 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
39_OHM_SE	*	Y	0.150 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
39_OHM_SE	ISL5,ISL8	Y	0.165 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
39_OHM_SE	TOP,BOTTOM	Y	0.175 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
42_OHM_SE	*	Y	0.130 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
42_OHM_SE	ISL5,ISL8	Y	0.145 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
42_OHM_SE	TOP,BOTTOM	Y	0.155 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE	*	Y	0.115 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
45_OHM_SE	ISL5,ISL8	Y	0.126 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
45_OHM_SE	TOP,BOTTOM	Y	0.135 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	*	Y	0.090 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
50_OHM_SE	ISL5,ISL8	Y	0.100 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
50_OHM_SE	TOP,BOTTOM	Y	0.105 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	*	Y	0.075 MM	0.075 MM	=STANDARD	=STANDARD	=STANDARD
55_OHM_SE	ISL5,ISL8	Y	0.080 MM	0.080 MM	=STANDARD	=STANDARD	=STANDARD
55_OHM_SE	TOP,BOTTOM	Y	0.085 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
68_OHM_DIFF	*	Y	0.171 MM	0.085 MM	=STANDARD	0.130 MM	0.1 MM
68_OHM_DIFF	TOP,BOTTOM	Y	0.185 MM	0.085 MM	=STANDARD	0.150 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	*	Y	0.136 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM
80_OHM_DIFF	TOP,BOTTOM	Y	0.141 MM	0.085 MM	=STANDARD	0.185 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	Y	0.121 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM
85_OHM_DIFF	TOP,BOTTOM	Y	0.125 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	Y	0.109 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM
90_OHM_DIFF	TOP,BOTTOM	Y	0.111 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	Y	0.086 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM
100_OHM_DIFF	TOP,BOTTOM	Y	0.090 MM	0.085 MM	=STANDARD	0.230 MM	0.1 MM

BGA

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BGA_P1MM	*	=STANDARD	?

Power and Common

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_ISO	*	=STANDARD	8000
GND_P2MM	*	=2:1_SPACING	1000
PWR_P2MM	*	=2:1_SPACING	1100

GENERIC

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GENERIC_ISO	*	=1:1_SPACING	?

BGA Area Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	BGA_P1MM

General Spacing Definitions

Default

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?

Fixed and Dielectric

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1:1_SPACING	*	0.1 MM	?
1X_DIELECTRIC	TOP,BOTTOM	0.071 MM	?
1X_DIELECTRIC	ISL3,ISL10	0.101 MM	?
1X_DIELECTRIC	*	0.076 MM	?

Board Stack-up

FINISHED BOARD THICKNESS: 1.94 MM

Top	Signal	0.5 oz (Cu plated)
	Prepreg	0.071 MM
2	Plane	1 oz
	Core	0.101 MM
3	Signal	0.5 oz
	Prepreg	0.115 MM
4	Plane	1 oz
	Core	0.076 MM
5	Signal	0.5 oz
	Prepreg	0.380 MM
6	Plane	1 oz
	Core	0.076 MM
7	Plane	1 oz
	Prepreg	0.380 MM
8	Signal	0.5 oz
	Core	0.076 MM
9	Plane	1 oz
	Prepreg	0.115 MM
10	Signal	0.5 oz
	Core	0.101 MM
11	Plane	1 oz
	Prepreg	0.071 MM
Btm	Signal	0.5 oz (Cu plated)

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